

ANALOG & DIGITAL CIRCUITS

Lecture 6

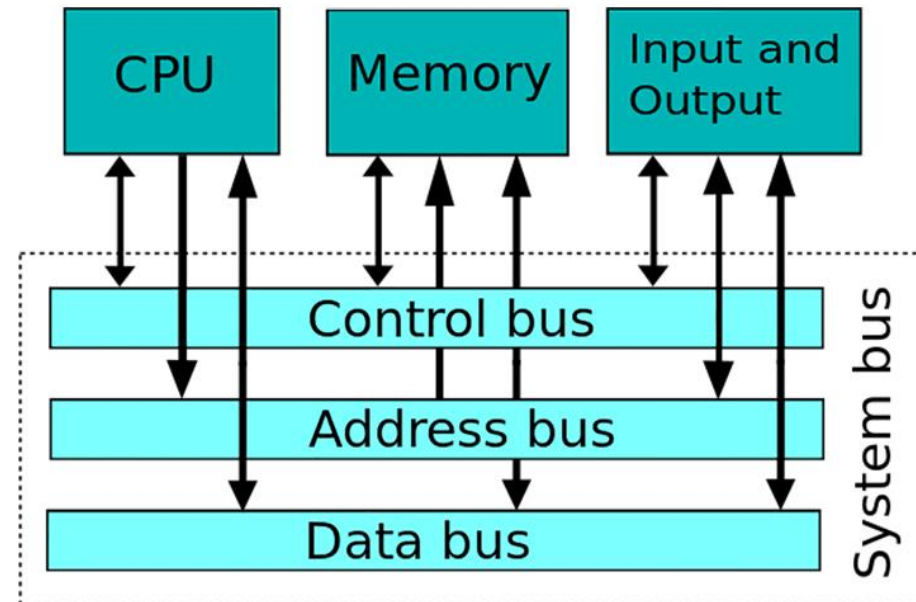
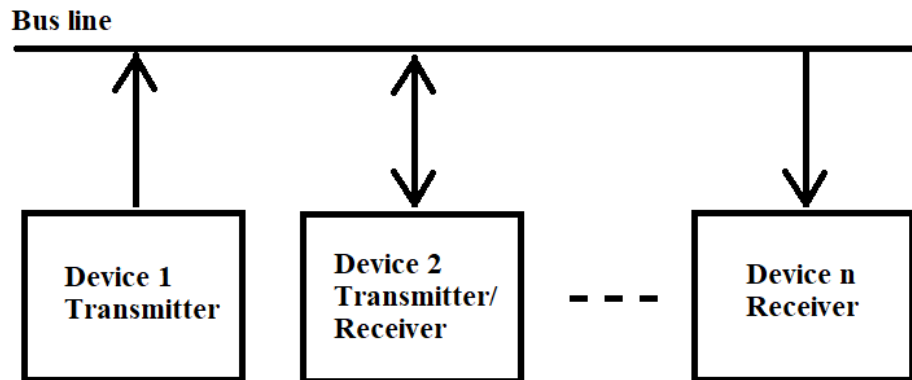
Computer Science Department,
TUCN

BUILDING A BUS USING LOGIC CIRCUITS

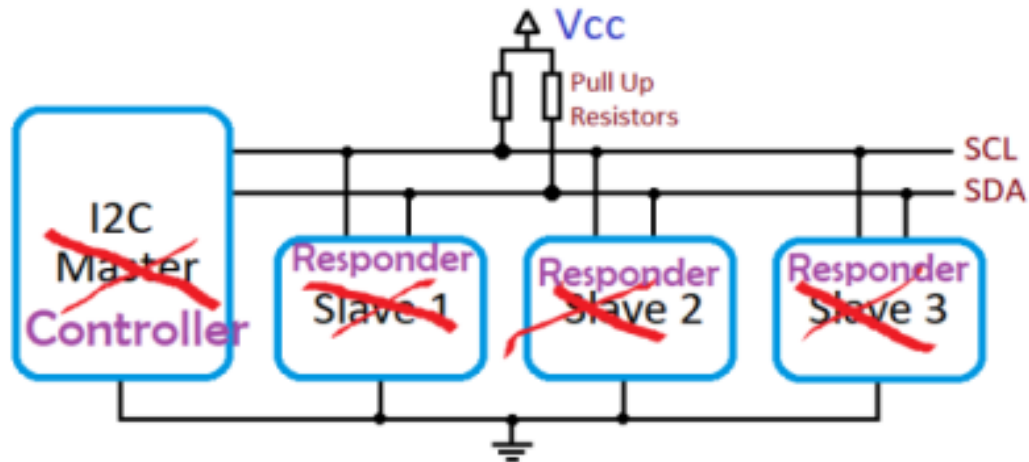
Introduction

Open Collector Logic Circuits

Three State Logic

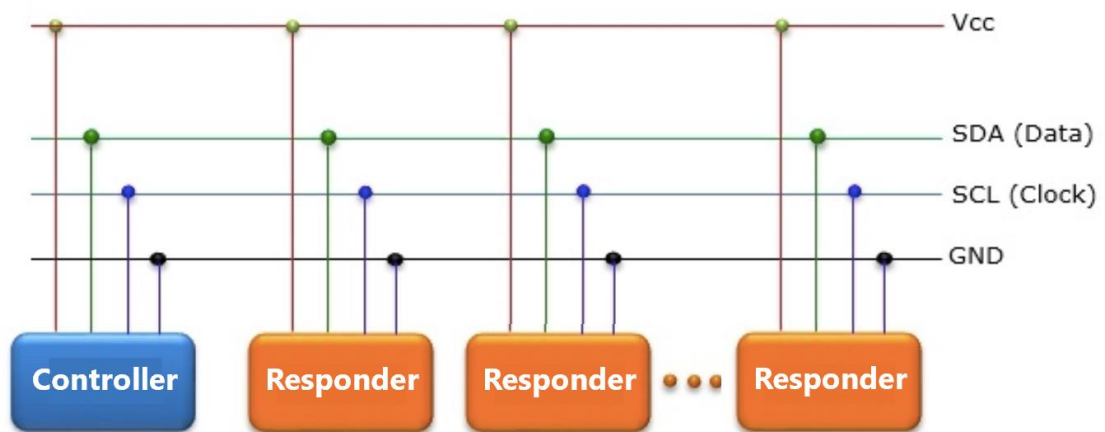


- SCL - Serial Clock Line
- SDA - Serial Data Line



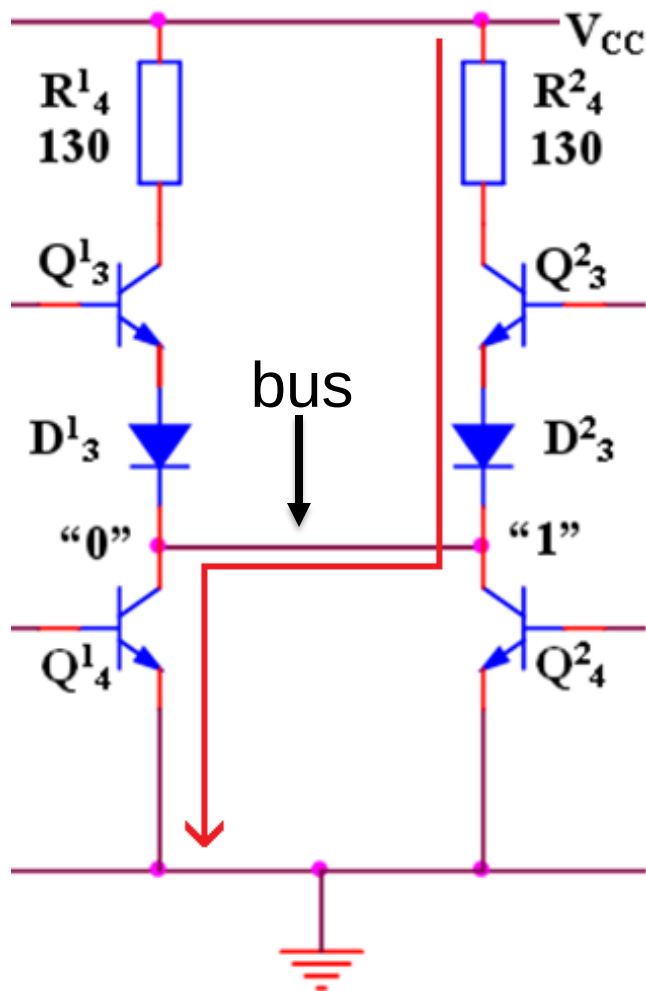
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- attach lower-speed peripheral ICs to processors and microcontrollers
- short-distance



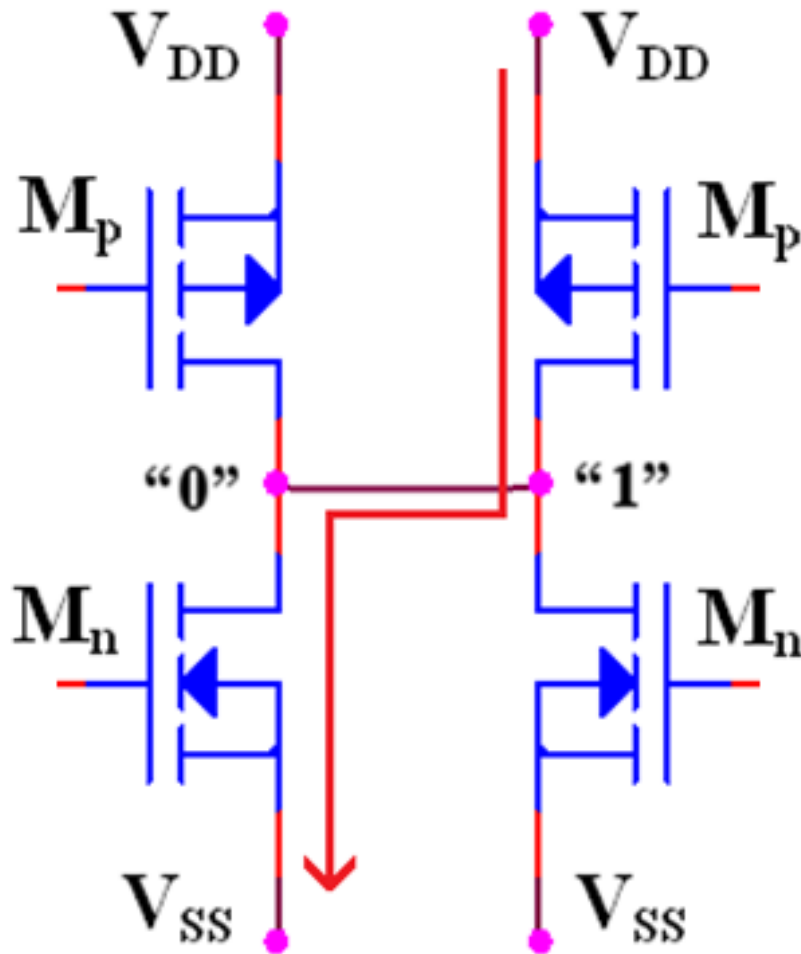
Examples

Introduction



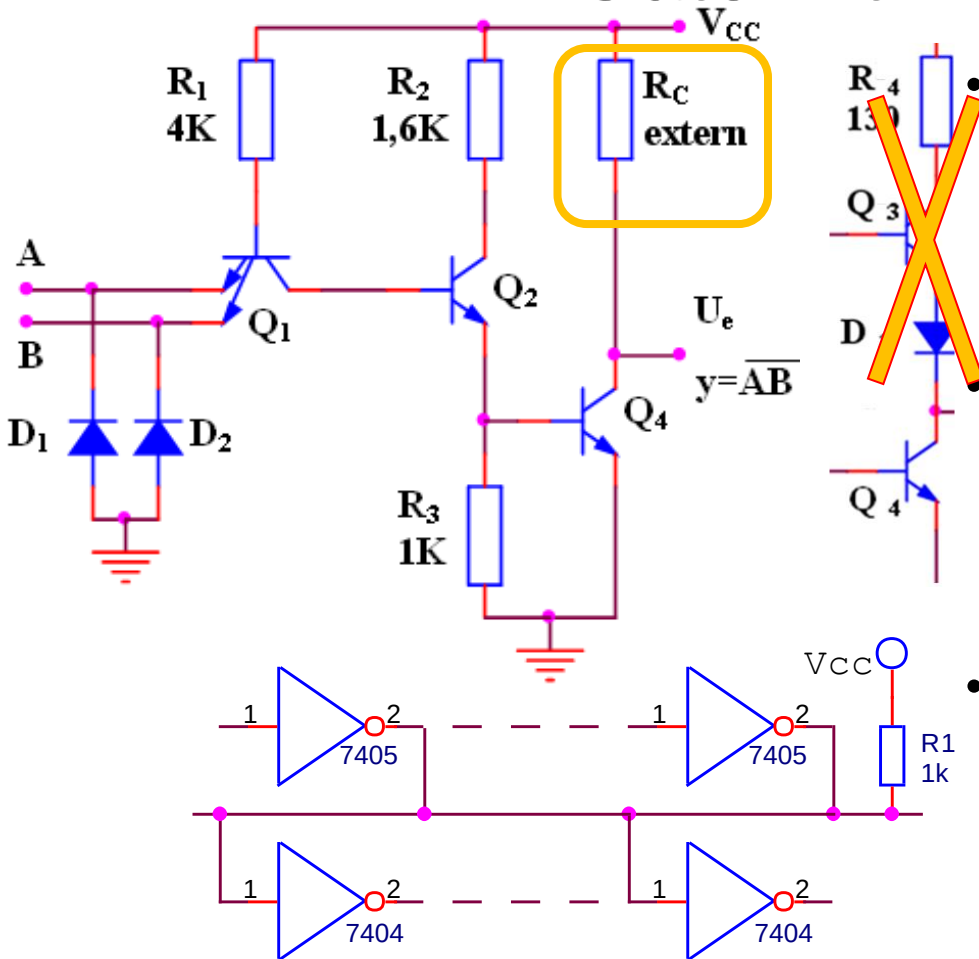
- Use of normal TTL or MOS logic circuits present drawback: to NOT allow the cabled AND function, i.e. the connection of all outputs together for achieving the AND logic function.
- Important issue for building up numeric systems: design of system buses
- See figure left: two standard TTL NAND gates have outputs connected together in parallel
- If both outputs are simultaneously “0” or “1” circuit is well operating

- If one output is “0” and the other “1”, practically transistor Q_3^2 is tied to ground, current through it being limited by diode D_3^2 and resistance R_4^2 (of 130 Ω). It involves a huge power dissipation on transistors Q_4^1 și Q_3^2 and through R_4^2 .
- Gate parameters are modifying or transistors are damaged by thermal excitement



- If outputs of two MOS gates are connected in parallel, for one output '1' and the other '0', the output voltage level is given by the voltage divider achieved by transistors with **n** channel and **p** channel being simultaneously in conduction
- For coupling more gates in parallel, special circuits are used, like:
 - **open collector circuits**
 - **open drain circuits**
 - **three state circuits**

TTL Gate with Open Collector



Open collector TTL gate has similar structure with standard TTL for input stage and medium stage (driving transistor Q_2); Output stage keeps only transistor Q_4

Collectors of Q_4 transistors from different gates may be tied together, the common connection being tied to power supply using a common external resistance

- Power supply voltage may present different values, not only 5V, making possible to generate at outputs other logic levels for '1'

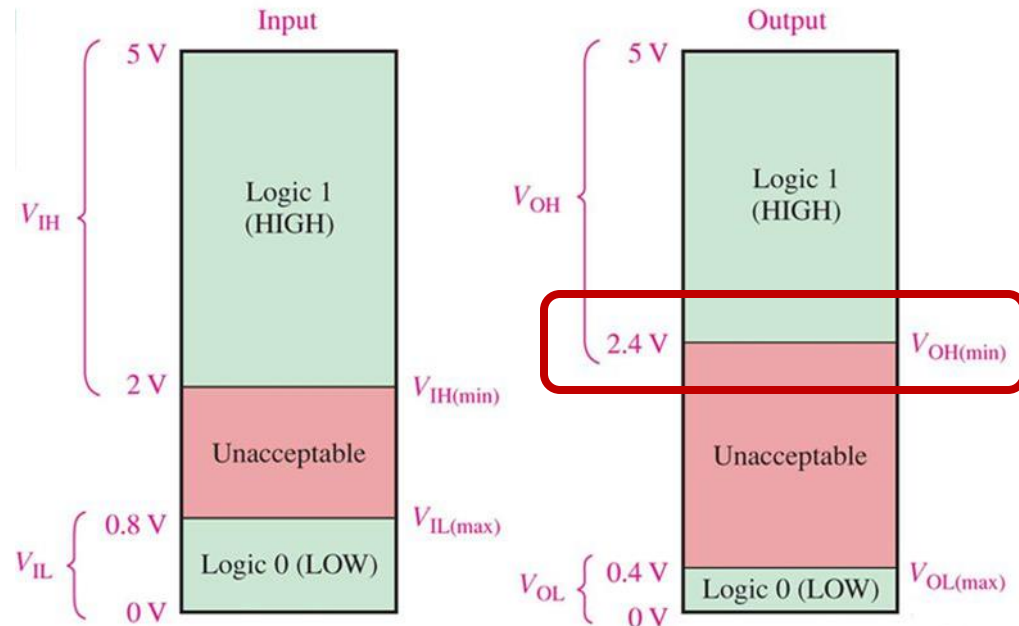
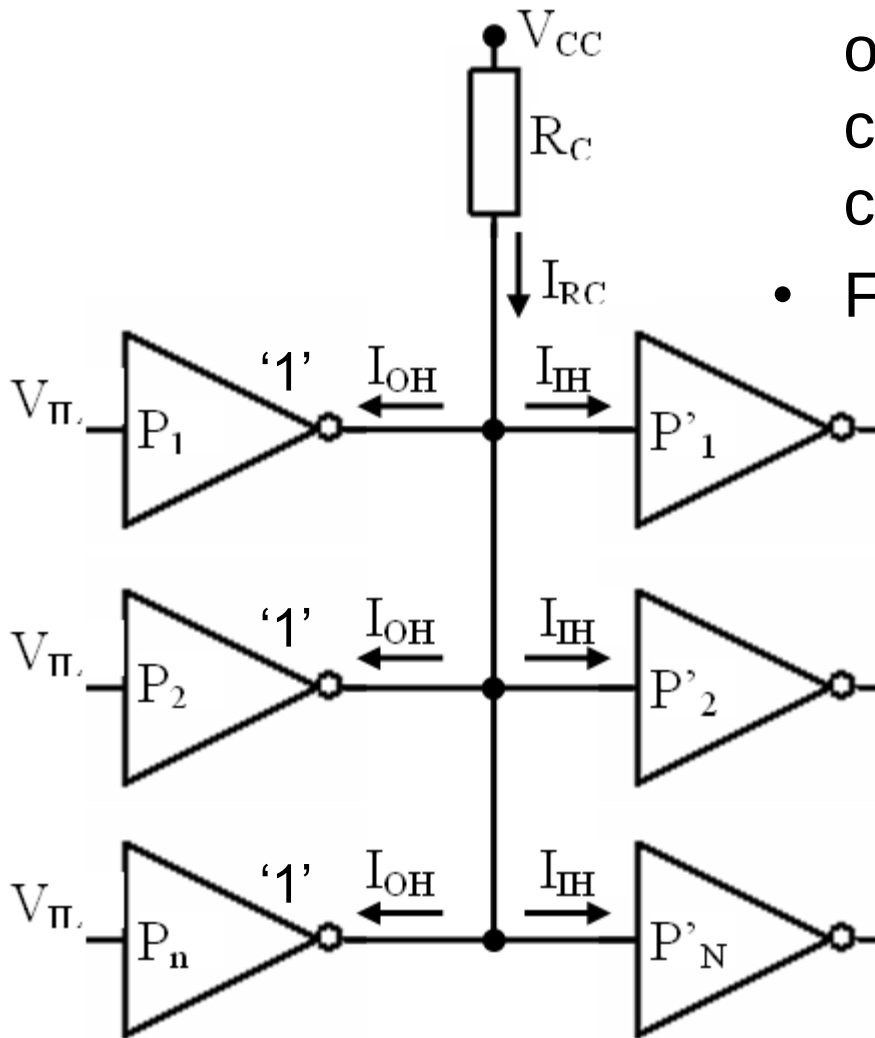
- Common external resistance R_C isn't an integrated one, but is designed as a function of number of TTL gates** connected together and number of inputs to be driven by this common output
- A great value for R_C reduces power dissipation but affects propagation delay and noise immunity
- There is a trade-off between designed power dissipation and delay

Calculus of R_C

Logic AND on the bus

- Function of logic level from common output, sunk (output) currents of connected gates and source (input) currents of driven gates
- For **'1' logic** R_C max will be:

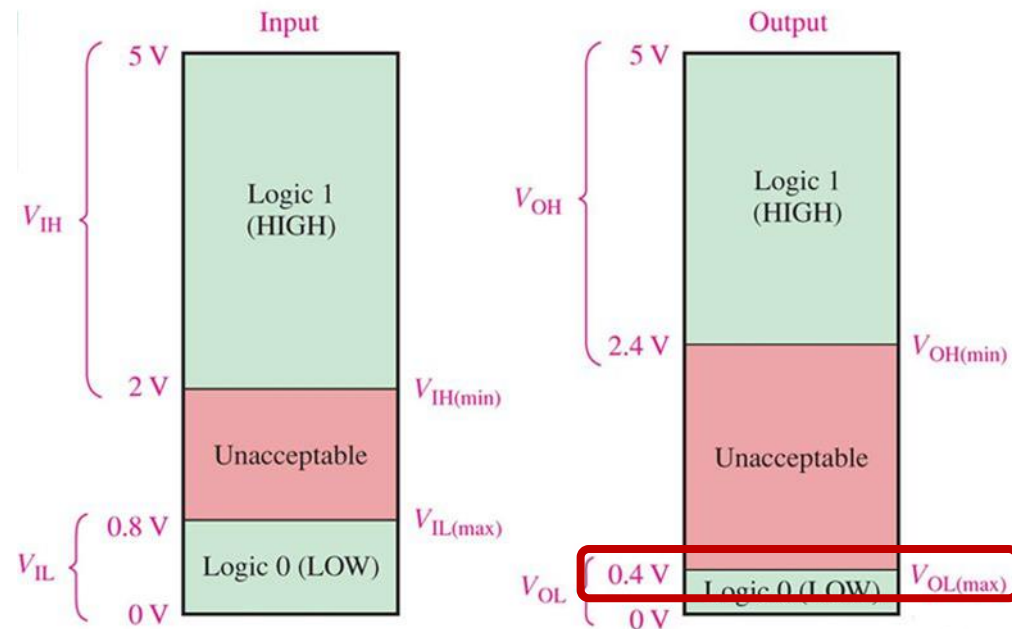
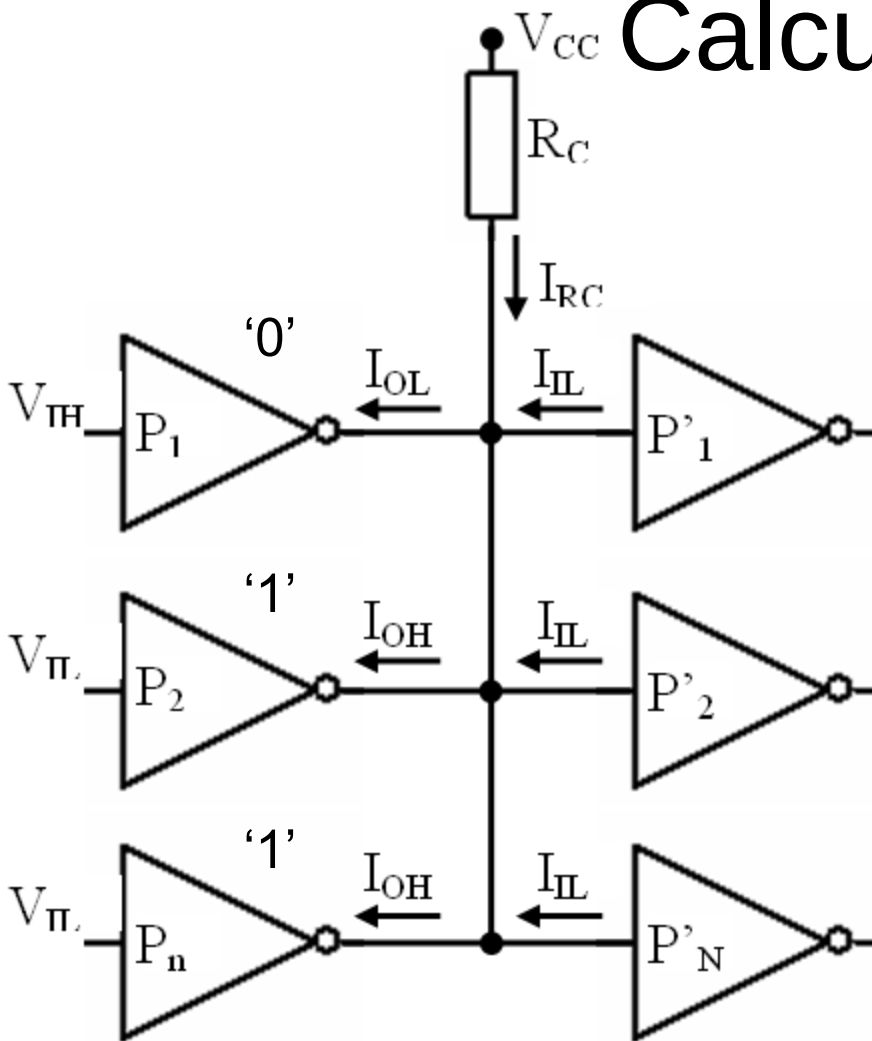
$$R_{C \max} = \frac{V_{CC \min} - V_{OH \min}}{n \bullet I_{OH} + N \bullet I_{IH}}$$



Calculus of R_C

- For '0' logic at output:

$$R_{C \min} = \frac{V_{CC \max} - V_{OL \max}}{I_{OL} + (n-1)I_{OH} - N \bullet I_{IL}}$$



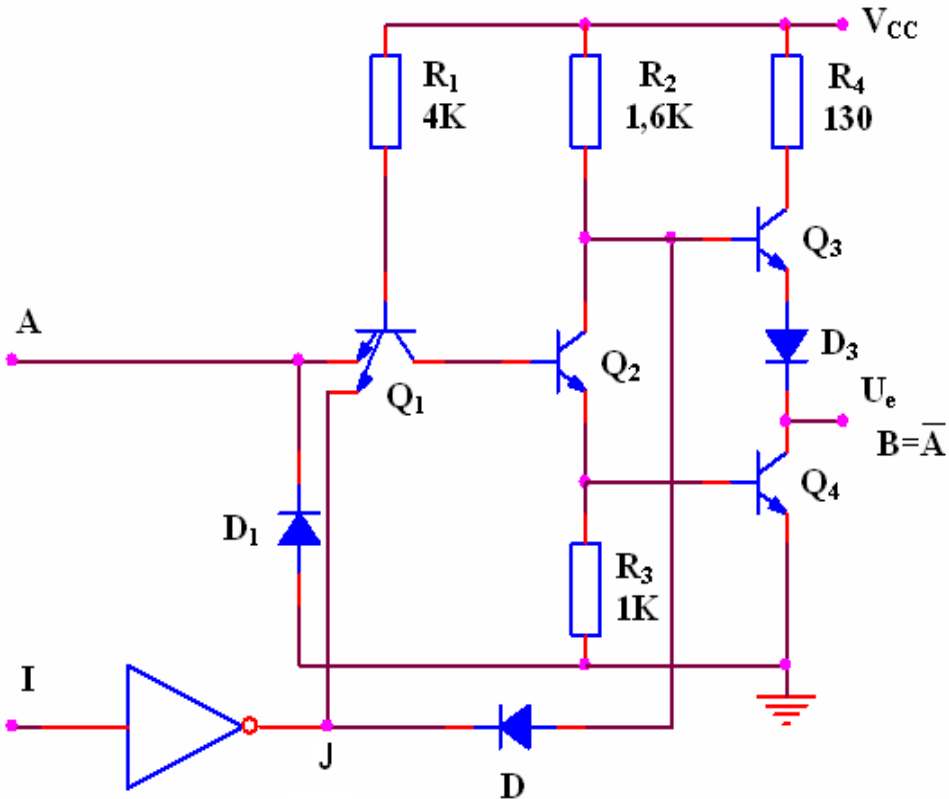
- $V_{CC} = 5V \pm 5\%$;
- $I_{OH} = 250\mu A$; $I_{OL} = 16mA$; $I_{IL} = 1.6mA$;
 $I_{IH} = 40\mu A$
- $P_D = 20mW$; $t_{pd} = 13ns$

How to choose the value of R_C ?

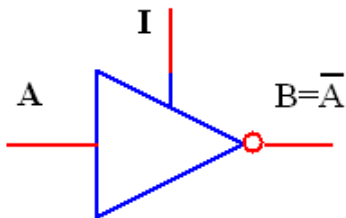
Three State TTL Logic Gates

- Both transistors of output circuit are locked
- Output circuit is separated
- From outside, TTL gate behaves as a high impedance
- Circuit presents **three states: “0”, “1” and high-impedance** state (state with both output transistors off)
- *The signal is left open, so that another output pin (e.g. elsewhere on a bus) can drive the signal or the signal level can be determined by a passive device (typically, a pull-up resistor).*

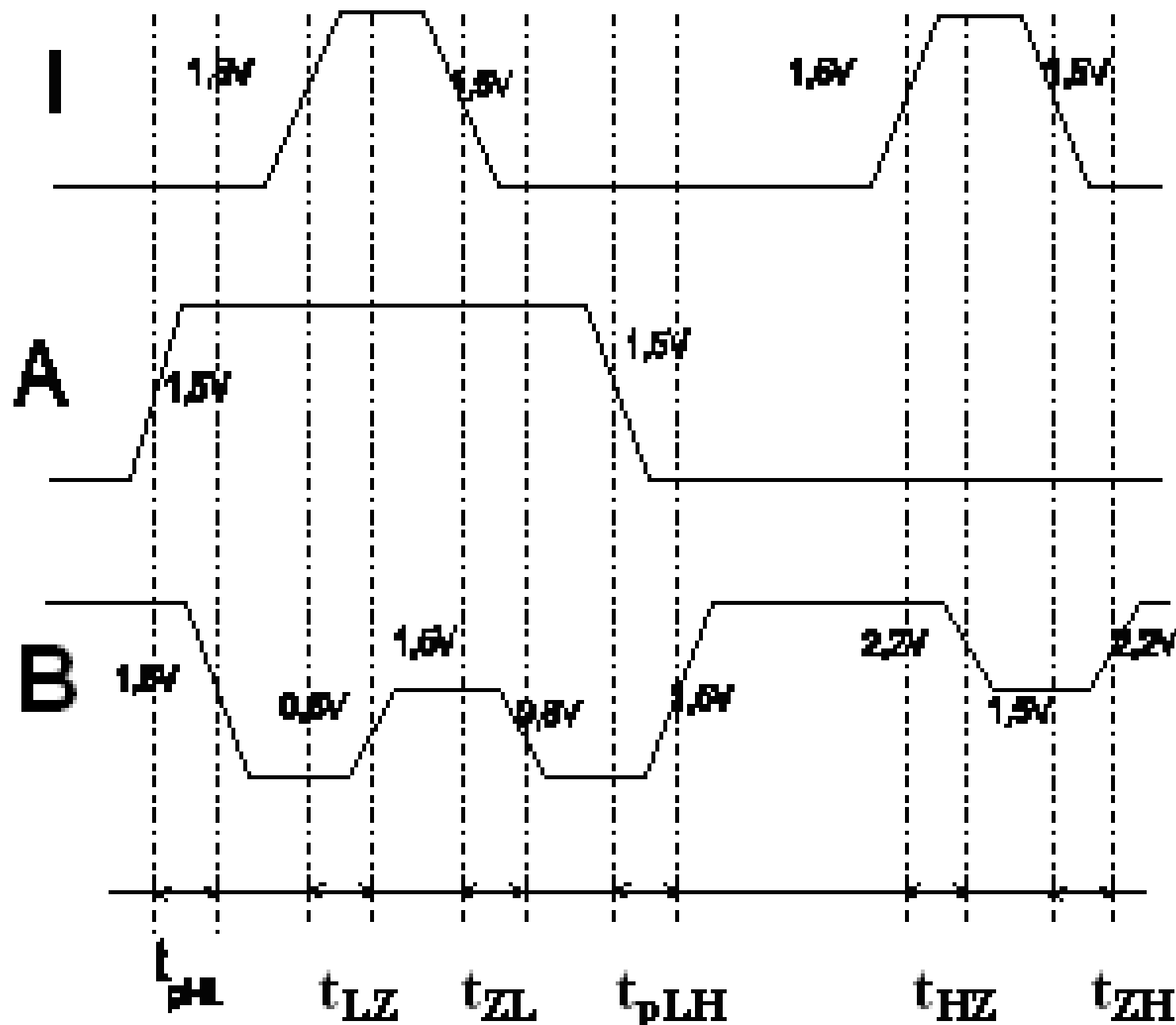
Three state Inverter



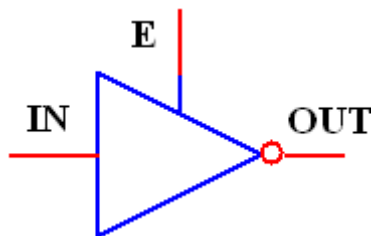
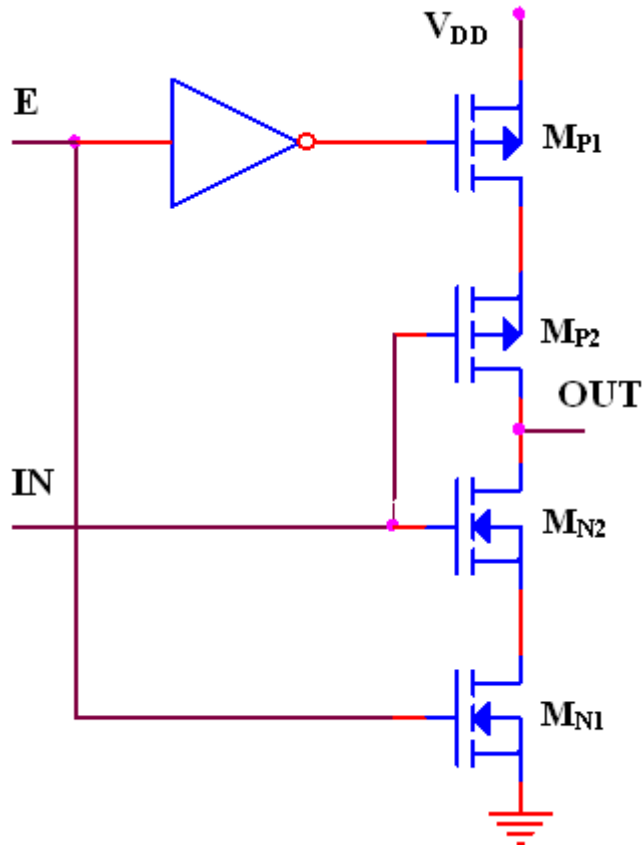
- **I="0"** – normal inverter, no influence
- **I="1"**, J="0", D open, Q₁ saturated, Q₂ and Q₄ off, Q₃ off, because through open diode D its base potential drops at 0.7V
- Circuit will present at output a state of high impedance (HZ state)
- During dynamic regime, besides t_{pLH} and t_{pHL} there are more parameters:
- Establishment of HZ from "0", t_{LZ} , and from "1", t_{HZ}
- Get-out times from HZ to "0", t_{ZL} , respectively in "1", t_{ZH}



- Total propagation delay is approx. 25 ns: greater than standard TTL, but less than TTL open collector
- $P_D=16mW$



Three state MOS inverter

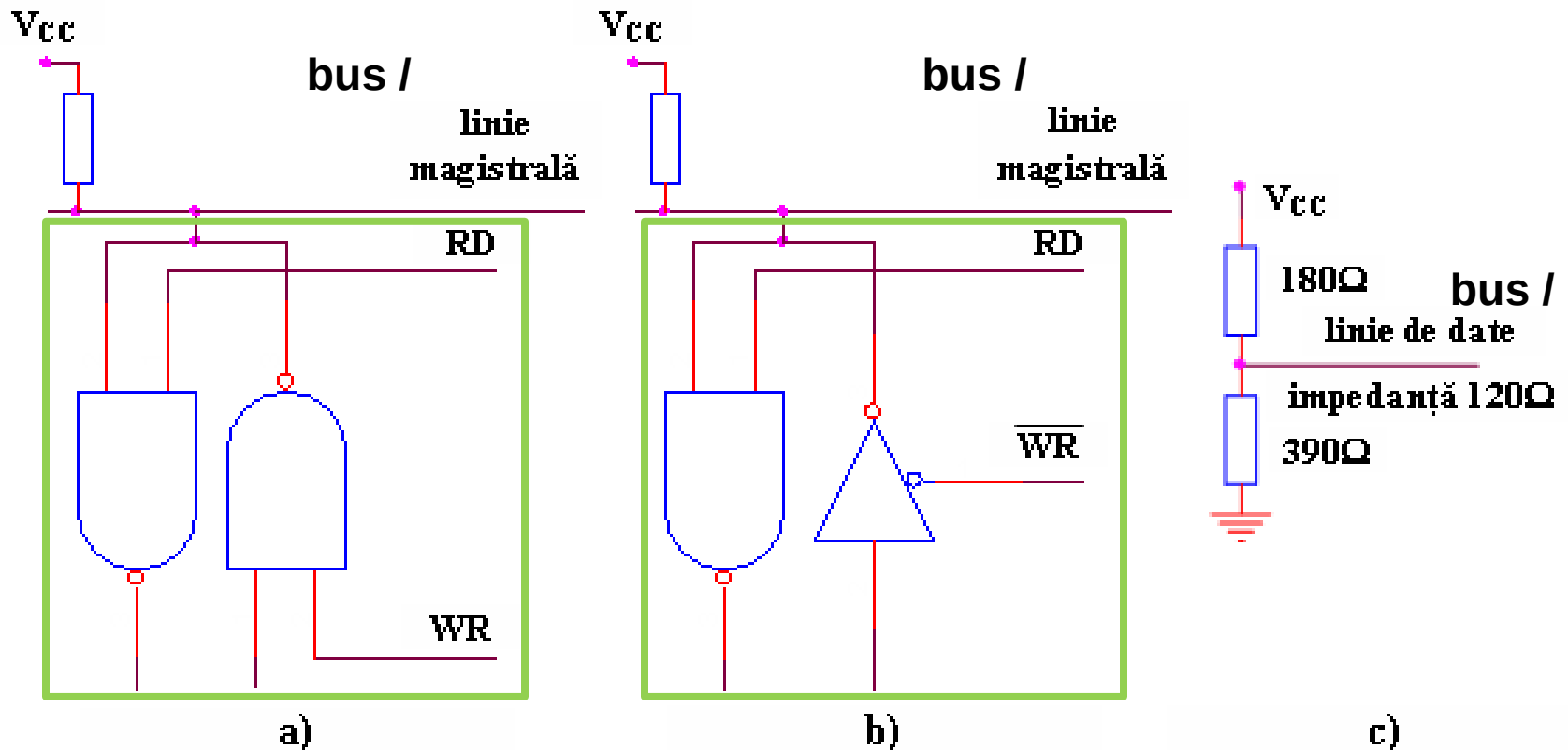


- Two **n** channel transistors and two **p** channel transistors
- One pair p-n transistors (denoted with 2) operates as standard inverter
- Second pair (denoted with 1) behaves as a switch (on/off), driven by input **E (enable)**
- If input **E is “1”**, M_{N1} and M_{P1} are open and circuit output presents logic level given by normal input IN
- If input **E is “0”**, M_{N1} and M_{P1} are off, and, no matter the input logic levels, the output presents high impedance state (higher than $10^{10}\Omega$ at 25°C).

Connecting circuits to buses

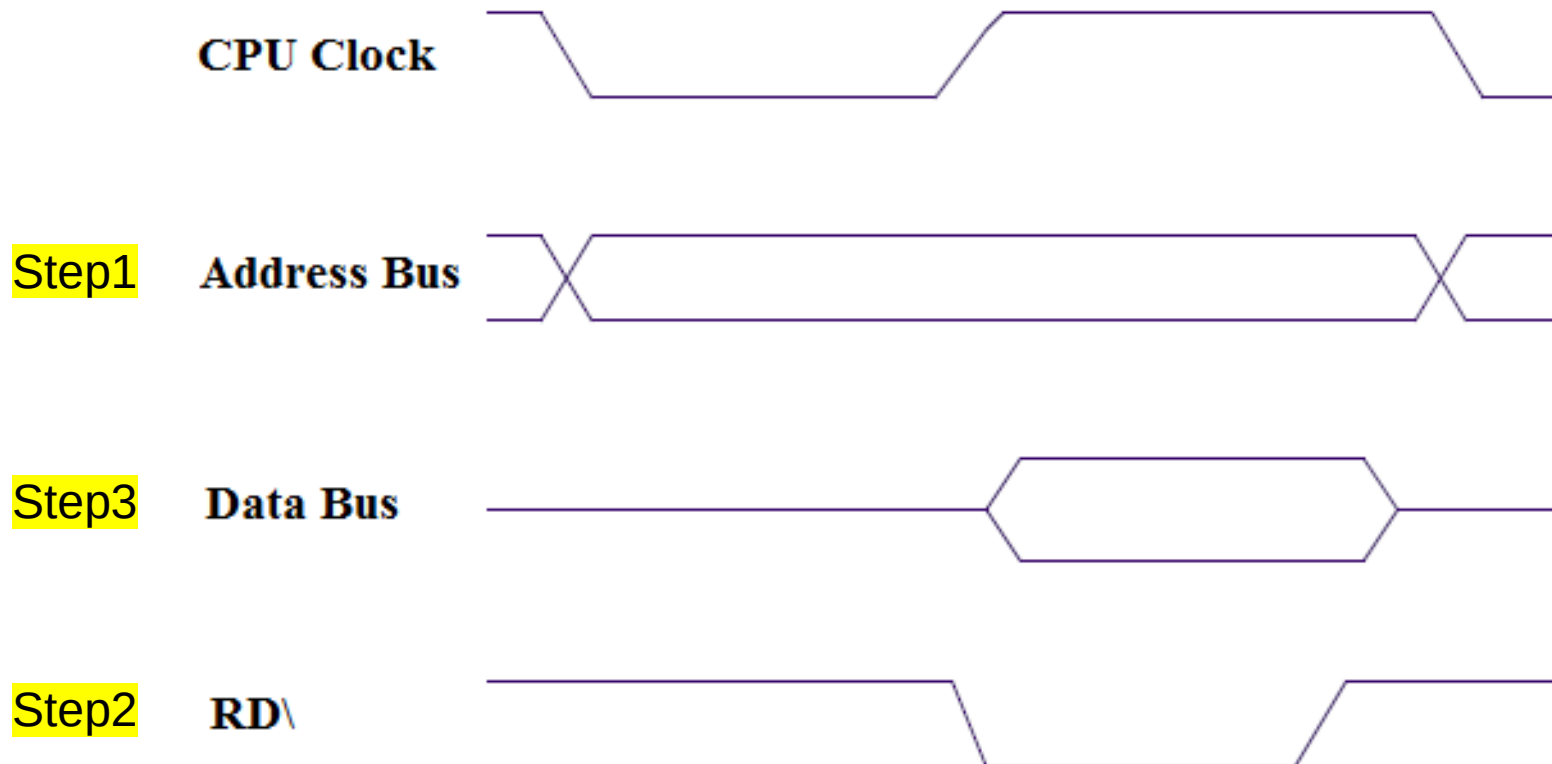
- A circuit connected at a bus presents a behavior of both receiver or sender (or is said that circuit is reading information from bus, and is writing info on that bus)
- There is need for **command signals: RD** and **WR**
 - the word on the bus is entered into the circuit via an RD\ control signal
 - by a WR\ control signal the word in the circuit is written on the bus

- When **bus is idle** (no activity) all connected lines are tied through a **pull-up resistance** to a high potential (usual power supply).
- If only **TTL lines** connected, instead of pull up resistance there are used **bus terminators** (mounted at end of bus, to avoid reflections)



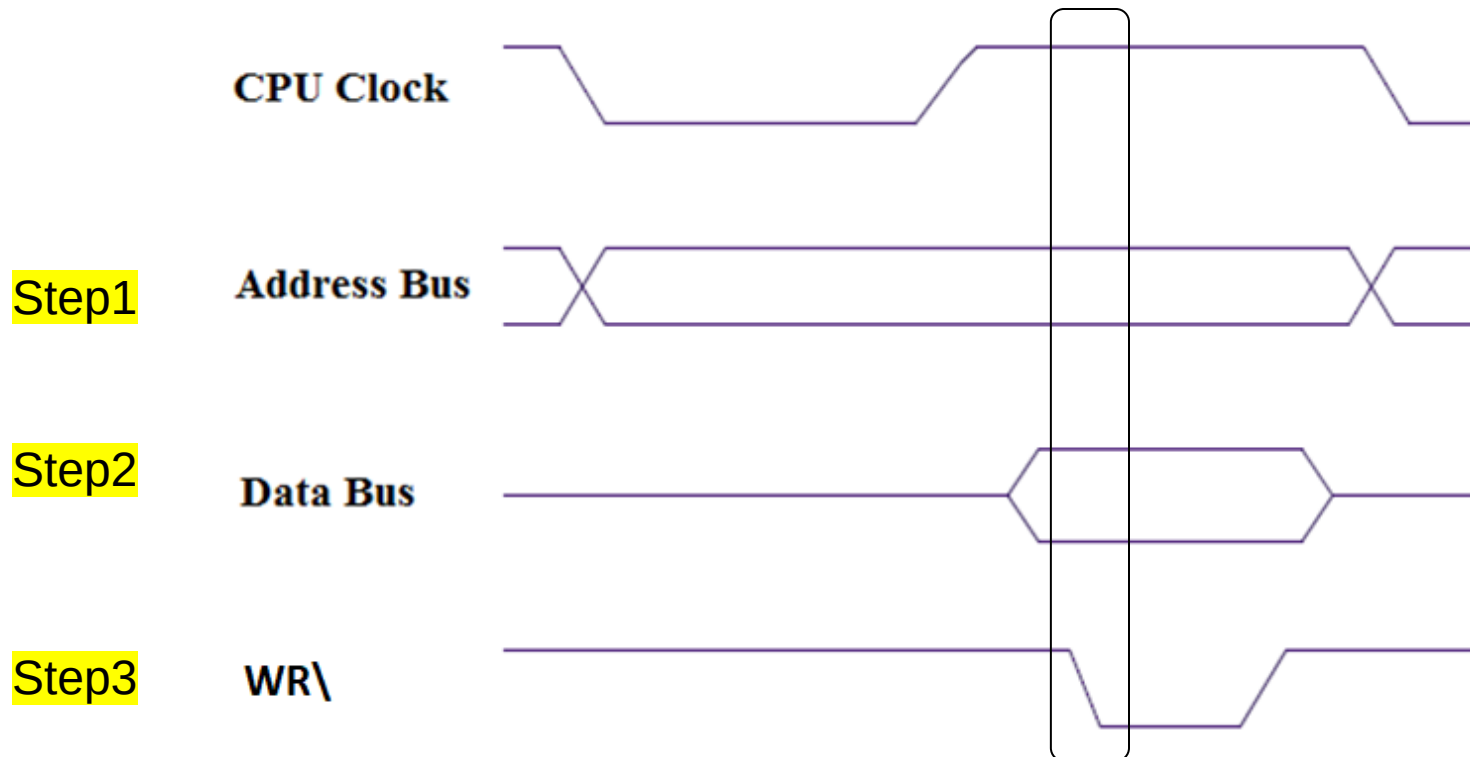
Example of CPU reading cycle

- CPU transmits the address on the address bus
- The RD\ control signal is switched to logic 0 by the CPU to signal to the external device that it is ready to take the data from the data bus
- CPU reads the data bus



Example of CPU write cycle

- CPU transmits the address on the address bus
- CPU provides data on the data bus
- The WR \ control signal is switched to logic 0 by the CPU to signal to the external device that the data is ready to be retrieved from the data bus
- The external device retrieves the data from the data store



TRANSFER BETWEEN REGISTERS and THREE STATE LOGIC

- Bus design is made using **Three State Logic**, implemented at circuit outputs
- For **information transfer**, *usually* are used three state **registers**, three state bus **amplifiers** (uni or bi directional), together with **decoders**

Pin diagram of the 374 (8-bit D-type flip-flop). The chip is shown with 20 pins. The central label is 374. The pin functions are as follows:

Pin Number	Function
1	OE
2	Q ₀
3	D ₀
4	D ₁
5	Q ₁
6	Q ₂
7	D ₂
8	D ₃
9	Q ₃
10	GND
11	CP
12	Q ₄
13	D ₄
14	D ₅
15	Q ₅
16	Q ₆
17	D ₆
18	D ₇
19	Q ₇
20	V _{CC}

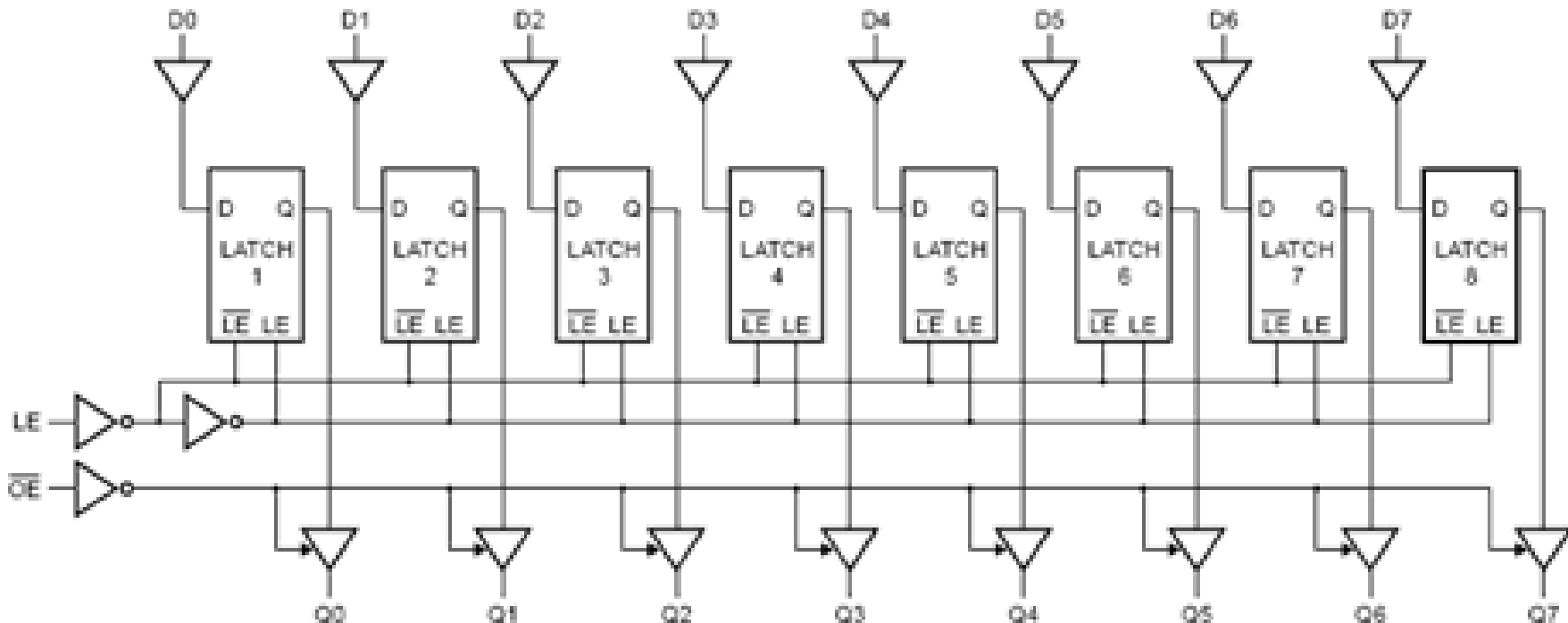
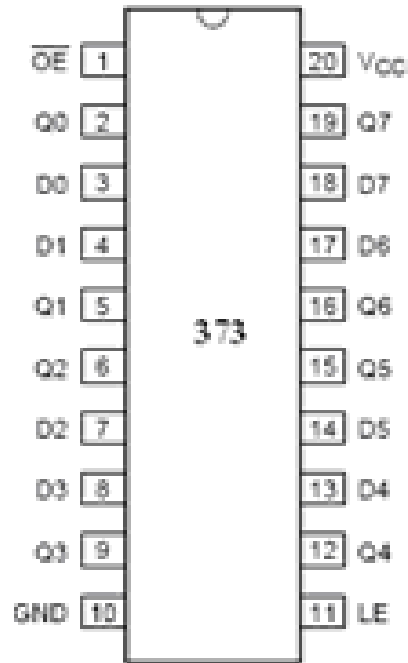
- If $OE \setminus = 0$

else

[illegible]

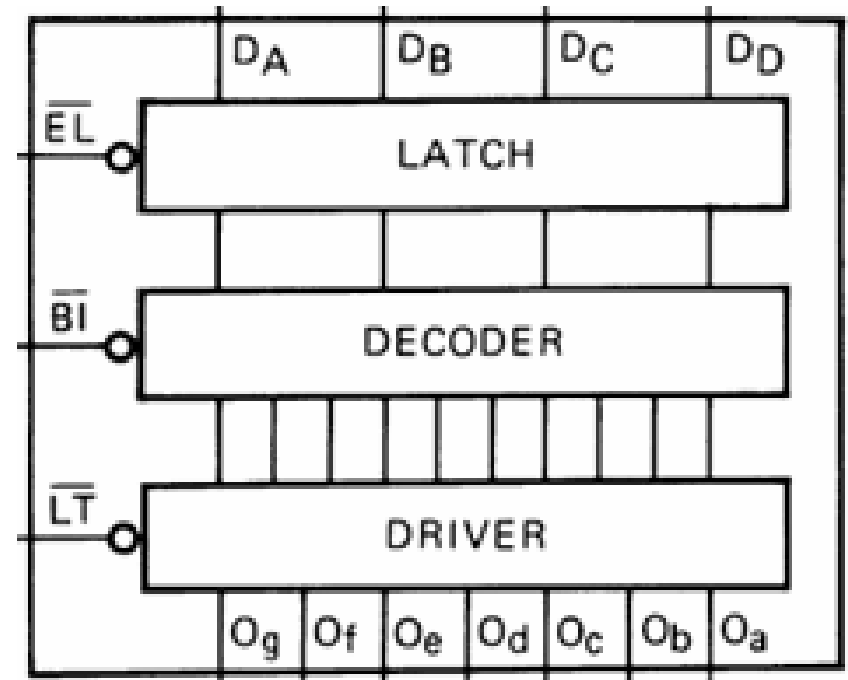
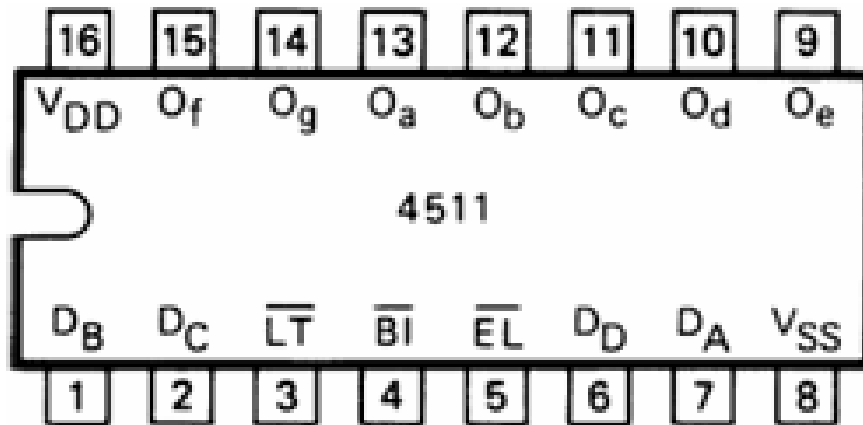
74373 Register

- 8 D type flip-flops, with outputs buffered with three-state circuits, enabled with OE\ input (OE\ zero-active)
- The data will be valid at the output only if OE \ is '0'.
- Data are sent at outputs while LE input is '1'.
- When LE goes '0' data are stored in latches
=> the last info is store only



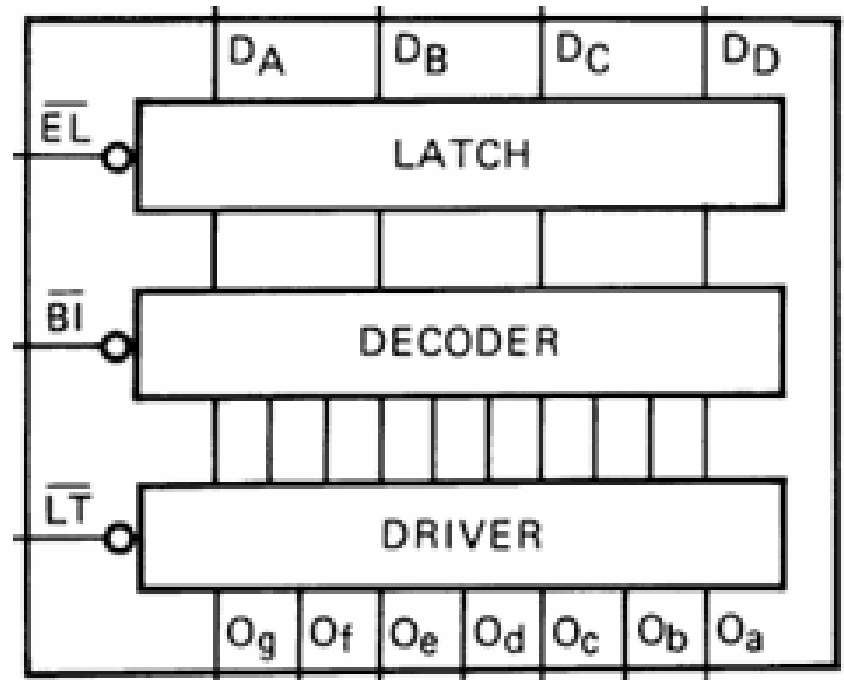
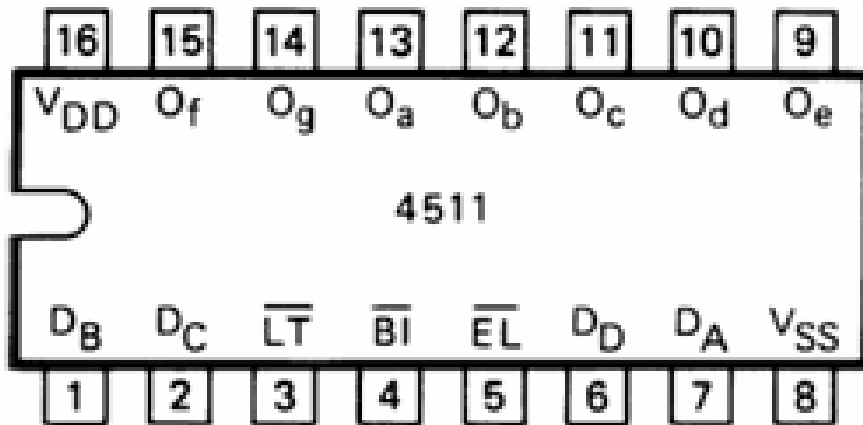
4511 - latch, decoder, driver

- Control of 7-segments LED display
- Roles:
 - Latch: store what I want to display
 - Decoder: from 4 bits – to 7 segments decoding
 - Driver: output the current needed for the LED

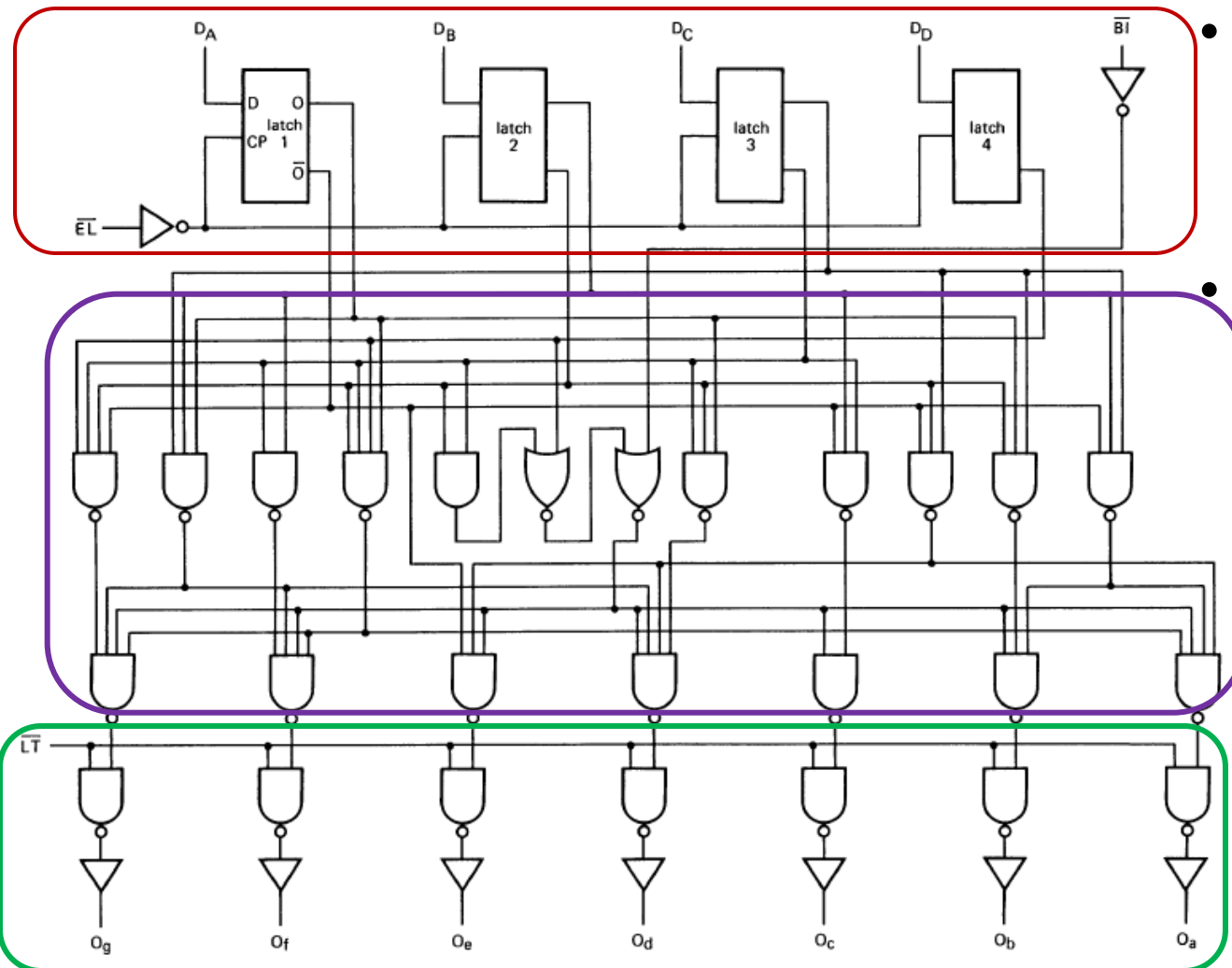


4511 - latch, decoder, driver

- Control of 7-segments LED display
- Functions: latch, decoder BCD to 7 segments and driver
- 4 data inputs (D_A - D_D)
- 1 latch enable input, '0' active ($\overline{EL}$)
- 1 blanking input (all segments off) , '0' active ($\overline{BI}$)
- 1 test lamp input (all segments on), '0' active ($\overline{LT}$)
- 7 outputs for segments, '1' active(O_a - O_g)



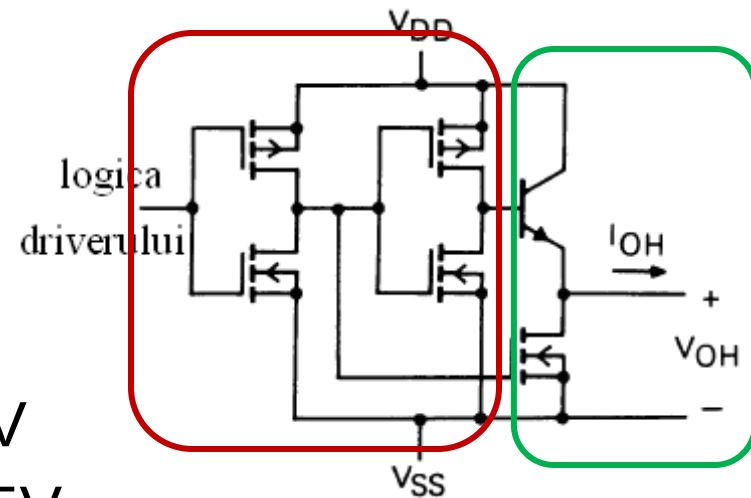
- When $\overline{EL}$ is '0', O_a-O_g are imposed by D_A-D_D
- When $\overline{EL}$ is '1', last values for D_A-D_D are latched and O_a-O_g present steady values
- If $\overline{LT}$ is '0', O_a-O_g are '1' no matter input values



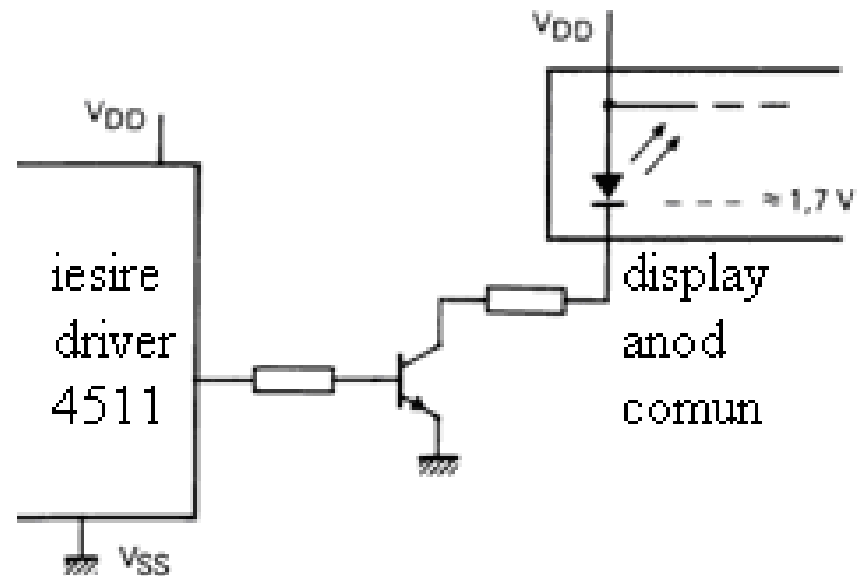
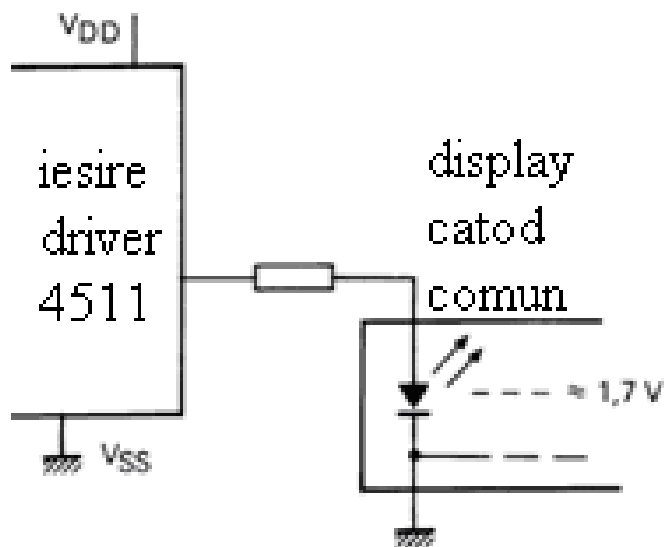
- If $\overline{LT}$ is '1', '0' value for $\overline{BI}$ drives O_a-O_g into '0'

- If $\overline{LT}$ and $\overline{BI}$ are '1', outputs O_a-O_g present values depending on decoder

- For having a high value for the output current, the output circuit is a bipolar transistor
- Output voltage depends on the generated current
- For $V_{DD}=5V$:
- $I_{OH}=0mA \rightarrow V_{OHmin}=4,10V, V_{OHtyp}=4,40V$
- $I_{OH}=10mA \rightarrow V_{OHmin}=3,60V, V_{OHtyp}=4,25V$
- $I_{OH}=20mA \rightarrow V_{OHmin}=2,80V, V_{OHtyp}=4,20V$

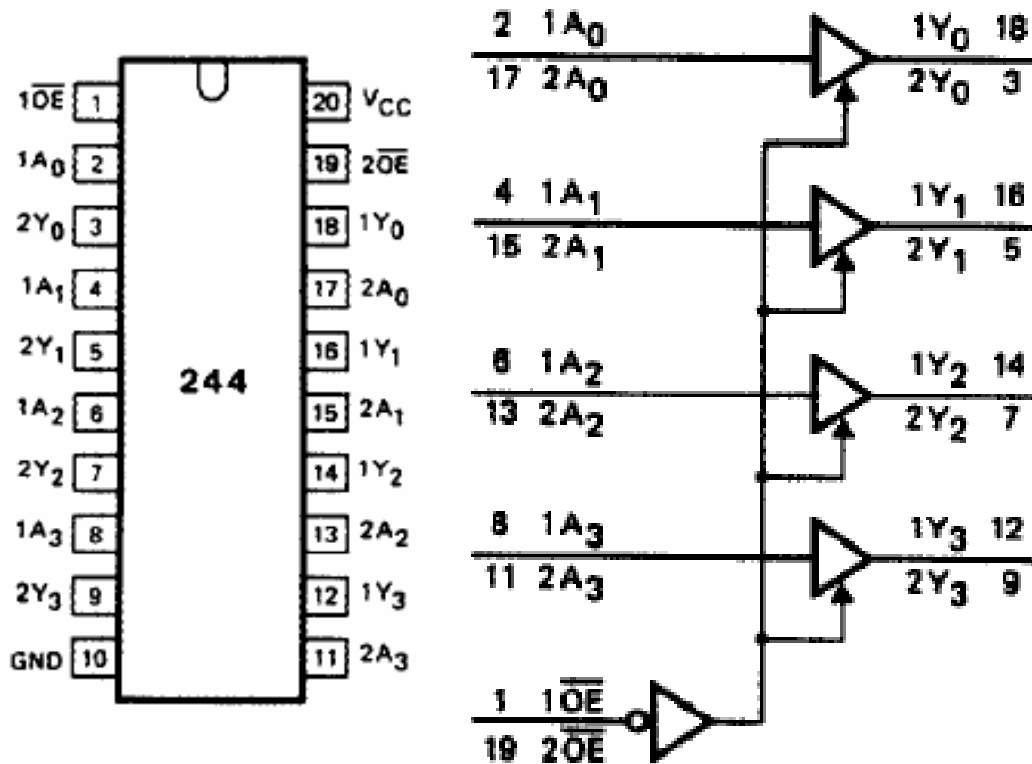


Red: logic done with MOS technology
Green: driver (current) with BJT



Unidirectional Amplifier 74244

- 8 three state amplifiers, outputs are effective if enable signals $1OE\backslash$ and $2OE\backslash$ are Low
- The two enable input signals allow to activate independently each set of 4 outputs: outputs $1Y_0 - 1Y_3$ will be enabled by low active signal $1OE\backslash$ and outputs $2Y_0 - 2Y_3$ will be enabled by signal $2OE\backslash$

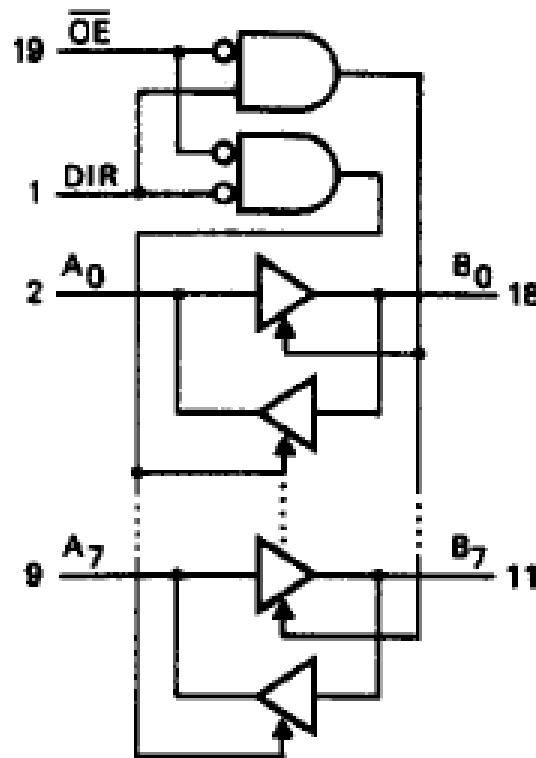
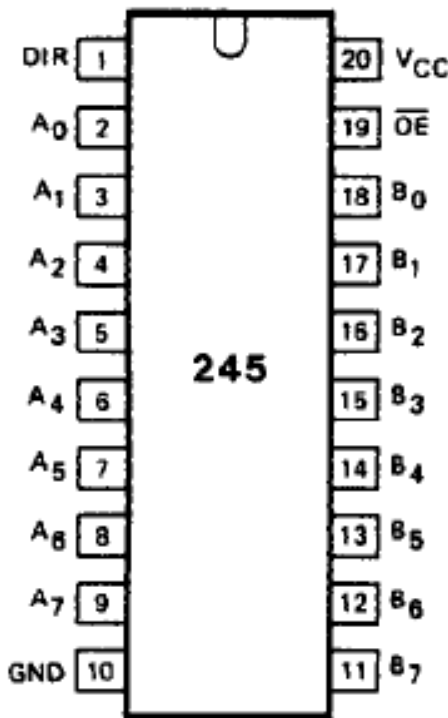


Digital amplifier:
if $In=0 \Rightarrow Out=0$
if $In=1 \Rightarrow Out=1$

Example: use on the address bus, where the transfer is unidirectional (we get address from the bus)

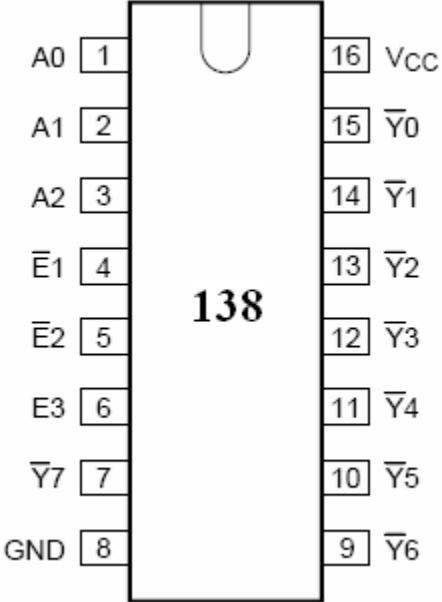
Bidirectional Amplifier 74245

- 16 three state amplifiers, eight for each direction
- Logic state for DIRECTION input signal DIR allows selecting group of eight amplifiers whose outputs may be activated (enabled) : **DIR='1'** data sent from inputs A_i to outputs B_i , **DIR='0'** data sent from inputs B_i to outputs A_i
- $\overline{OE}$ ='1' disables all outputs, $\overline{OE}$ ='0' enables outputs selected by signal DIR

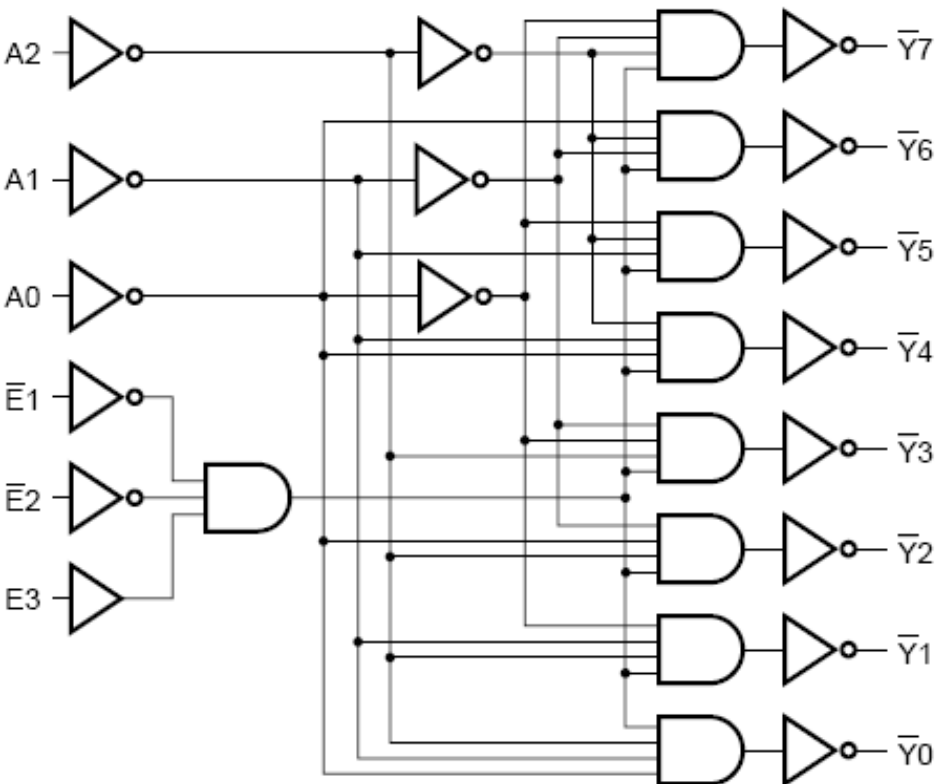


Example: use on the data bus, where the transfer is bidirectional (we put and get data on/from the bus)

Decoder 74138



- 3 address inputs, 8 Low active outputs and 3 enable signals
- If $E1 \neq '0'$, $E2 \neq '0'$ si $E3 = '1'$ output Y_i selected by binary combination of address signals $A0$, $A1$ & $A2$, will be '0', the rest of outputs being '1'



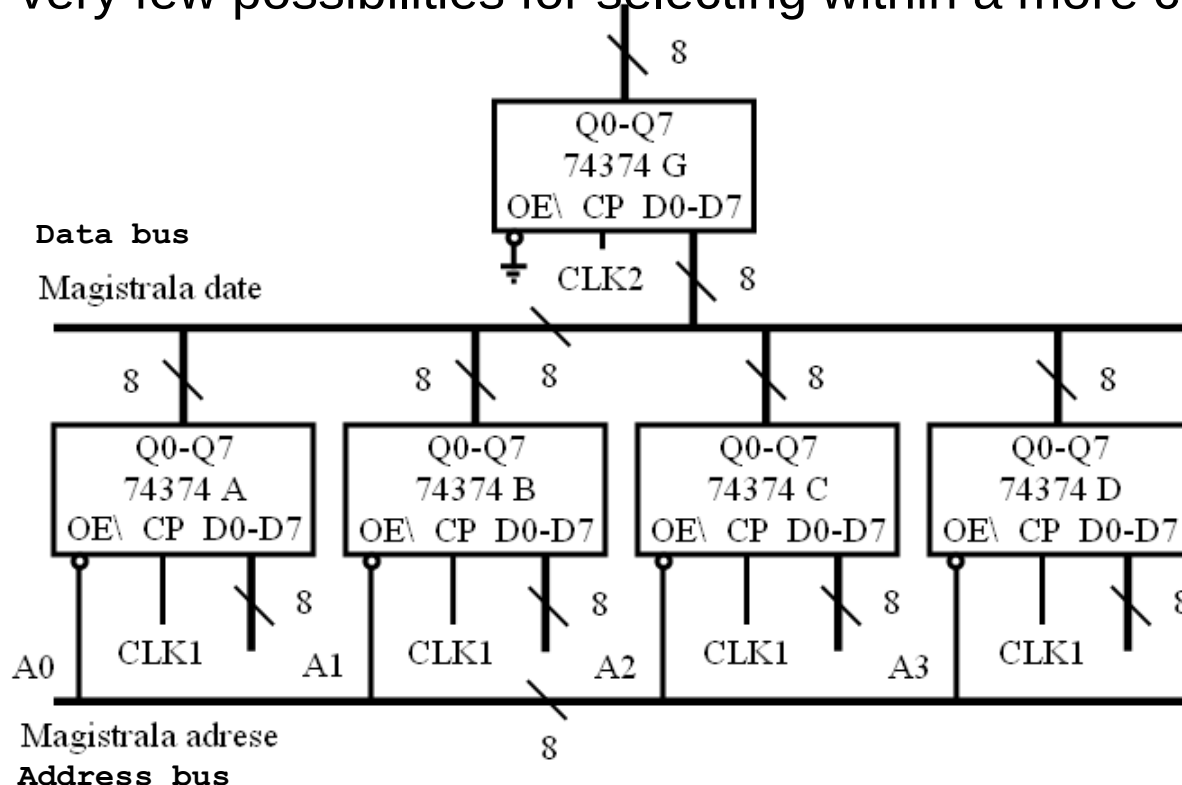
• For any other combination of values for enable inputs $E1$, $E2$ & $E3$, all outputs will be '1'

• For disabled circuit, all outputs will NOT be in HZ state, but in logic '1'; this way, the low active outputs will NOT select other circuits

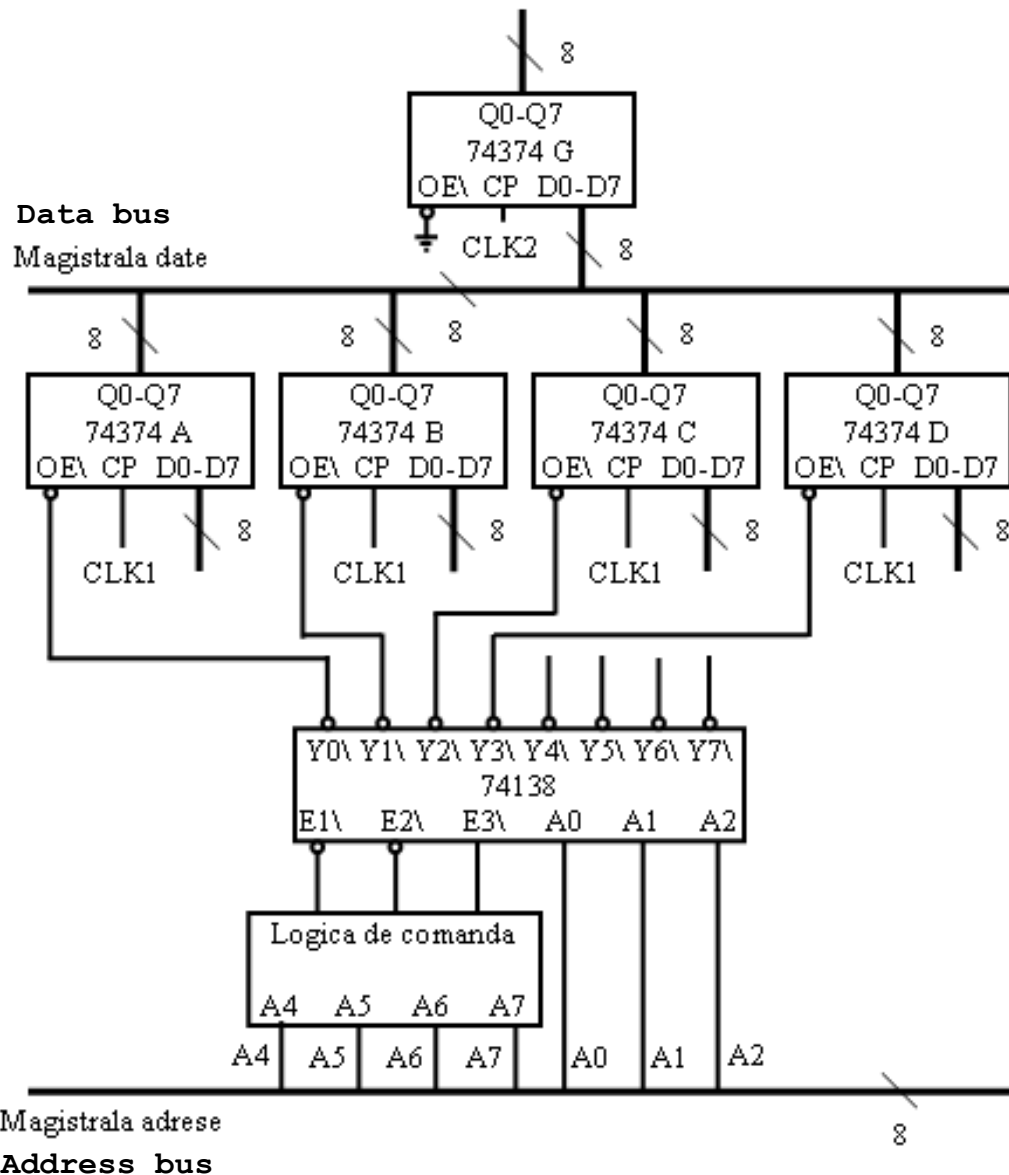
Example: use to receive an address and activate with $Y_i = 0$ only a circuit (from a range of circuits connected to a bus)

REGISTER TRANSFER (1)

- Data from one (out of four registers: A, B, C or D), is transferred in a fifth one, G
- Common data bus (8 lines), either output data from registers A, B, C or D, either input data for register G
- **Linear selection** method: only one address line used for selection (no decoder needed)
- Advantage – minimum complexity
- Drawback – very few possibilities for selecting within a more complex circuit



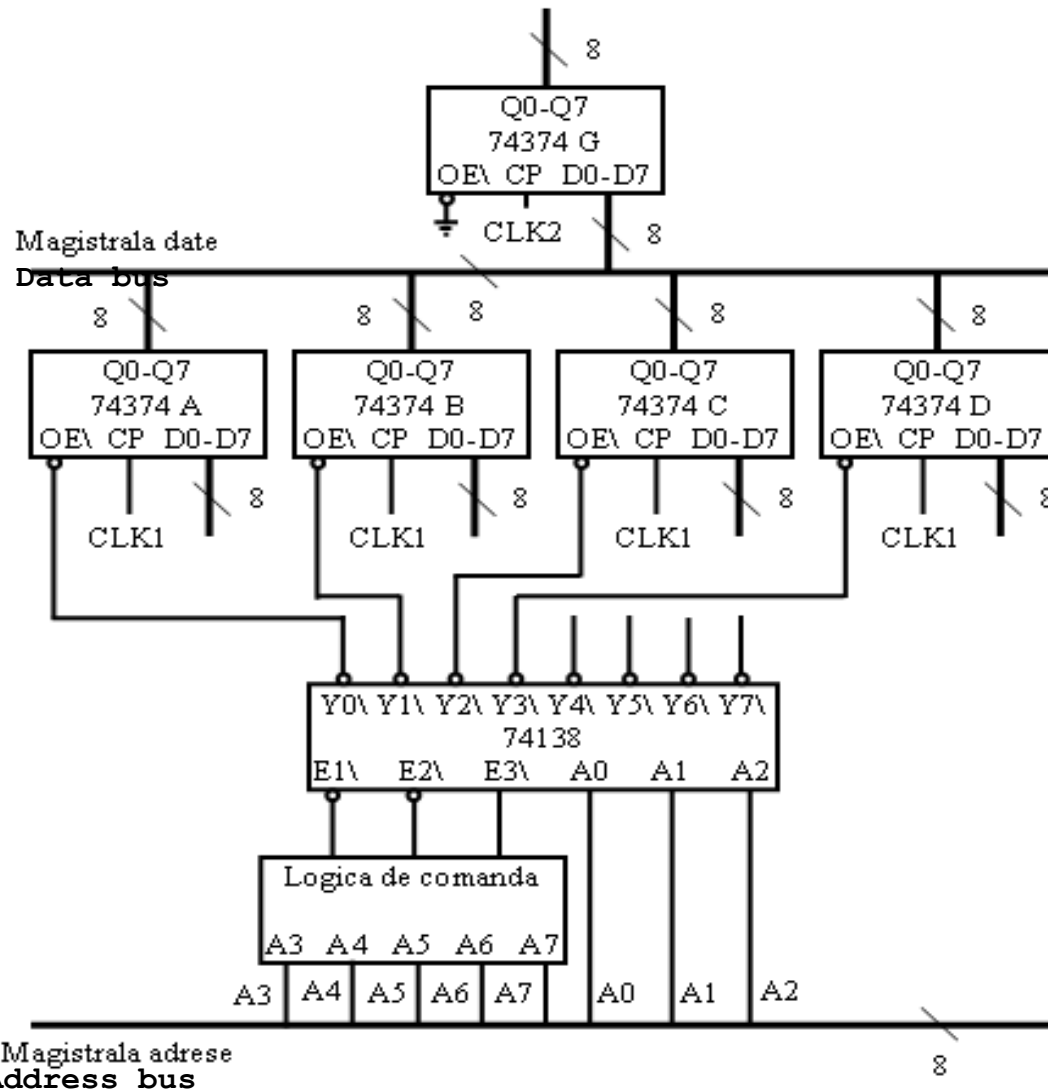
REGISTER TRANSFER (2)



- For **partial selection** method, only a part of address space is used
- Advantage – lower complexity
- Drawback – fewer selection possibilities

Example: A3 is missing from the address bus selection

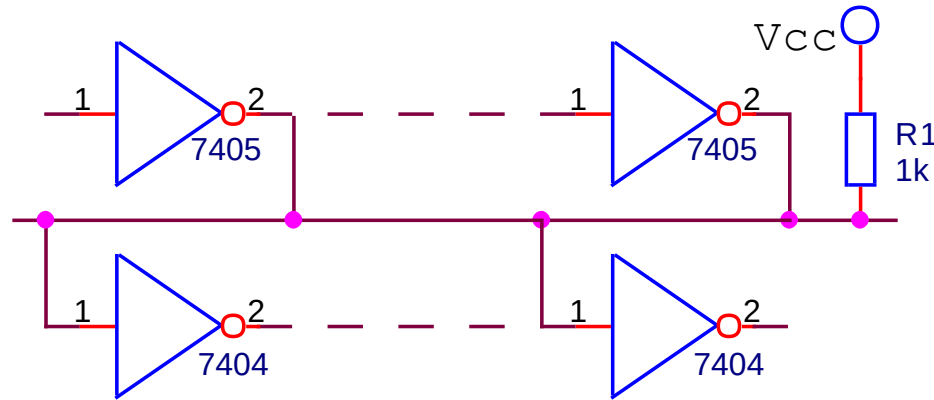
TRANSFER BETWEEN REGISTERS (3)



- Activating one register: a result of decoding some address lines from the address bus; corresponding active Output Enable line (OE\) will select register
- For selecting a register **all address lines** are considered (**total selection**)
- *Advantage* – all address space may be used (N addresses $\Rightarrow 2^N$ devices/ports)
- *Drawback* – selection circuit complex
- Data writing made on raising edges of the clock signals CLK1 & CLK2
- Register G is selected all time

Proposed Problems

- Design an un-adapted transmission line with 5 OpenCollector transmitters and 5 TTL standard receivers.



$$R_{1\max} = \frac{V_{CC\min} - V_{OH\min}}{5I_{OH} + 5I_{IH}} \approx 1,6k\Omega$$

$$R_{1\min} = \frac{V_{CC\max} - V_{OL\max}}{I_{OL} + 4I_{OH} - 5I_{IL}} \approx 0,5k\Omega$$

$$R_1 = 1k\Omega$$

Select R1 to an average value => good Noise Margins

- Design an adapted line with 10 OC transmitters and 2 TTL standard receivers. Bus line has a characteristic impedance of $Z_0 = 250\Omega$.

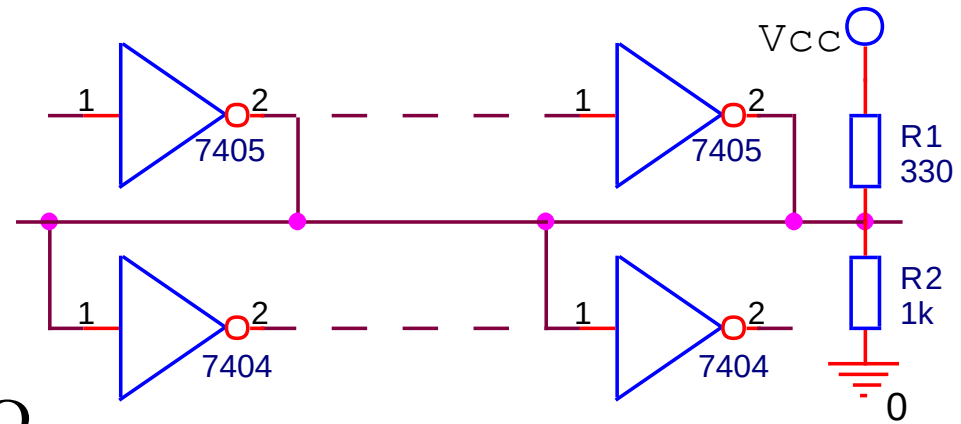
$$R_{1\max} = \frac{V_{CC\min} - V_{OH\min}}{10 I_{OH} + 2 I_{IH}} \approx 910\Omega$$

$$R_{1\min} = \frac{V_{CC\max} - V_{OL\max}}{I_{OL} + 9 I_{OH} - 2 I_{IL}} \approx 322\Omega$$

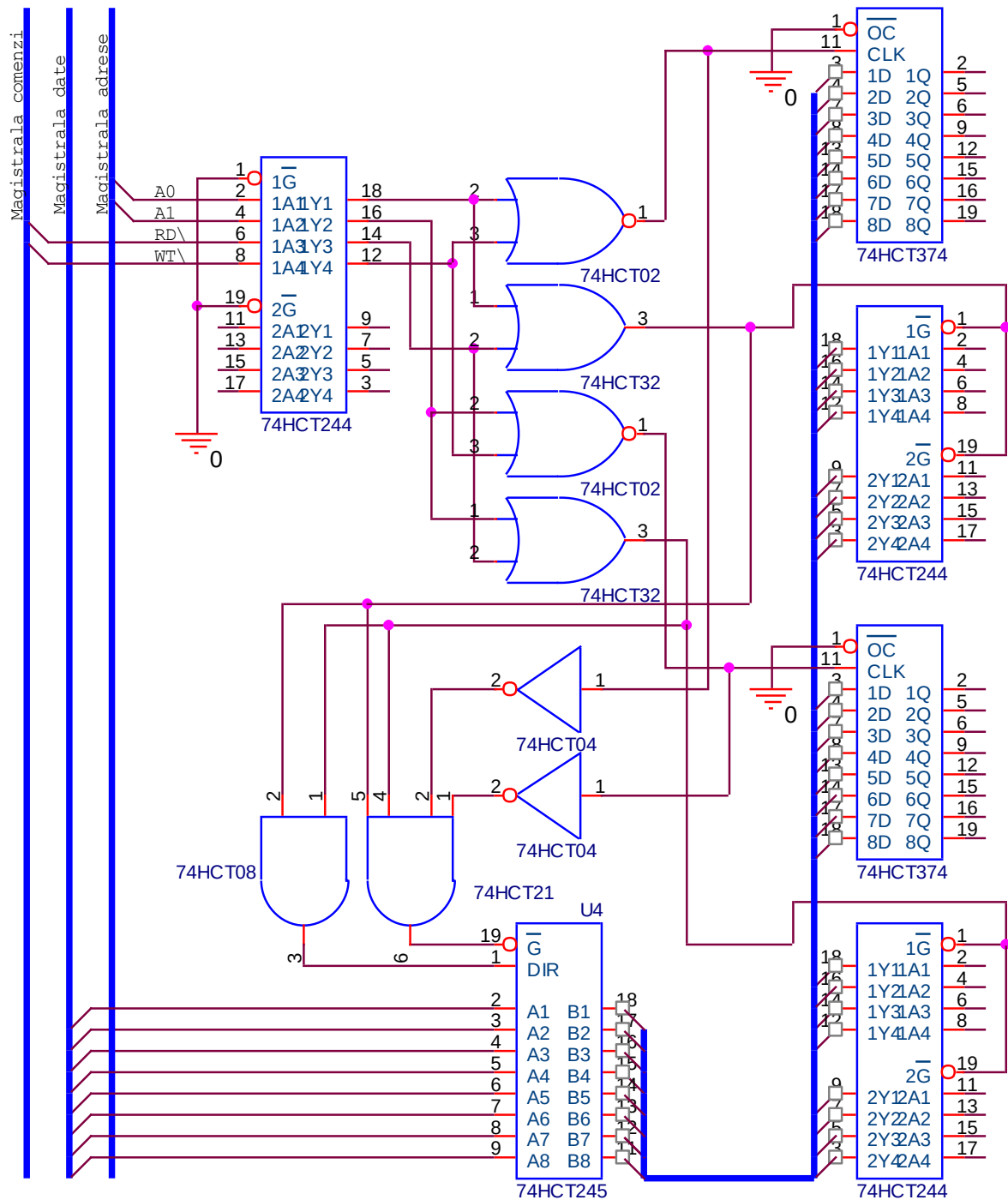
$$R_1 = 330\Omega$$

$$Z_0 = \frac{R_1 R_2}{R_1 + R_2}; R_2 = \frac{Z_0 R_1}{R_1 - Z_0} \approx 1k\Omega$$

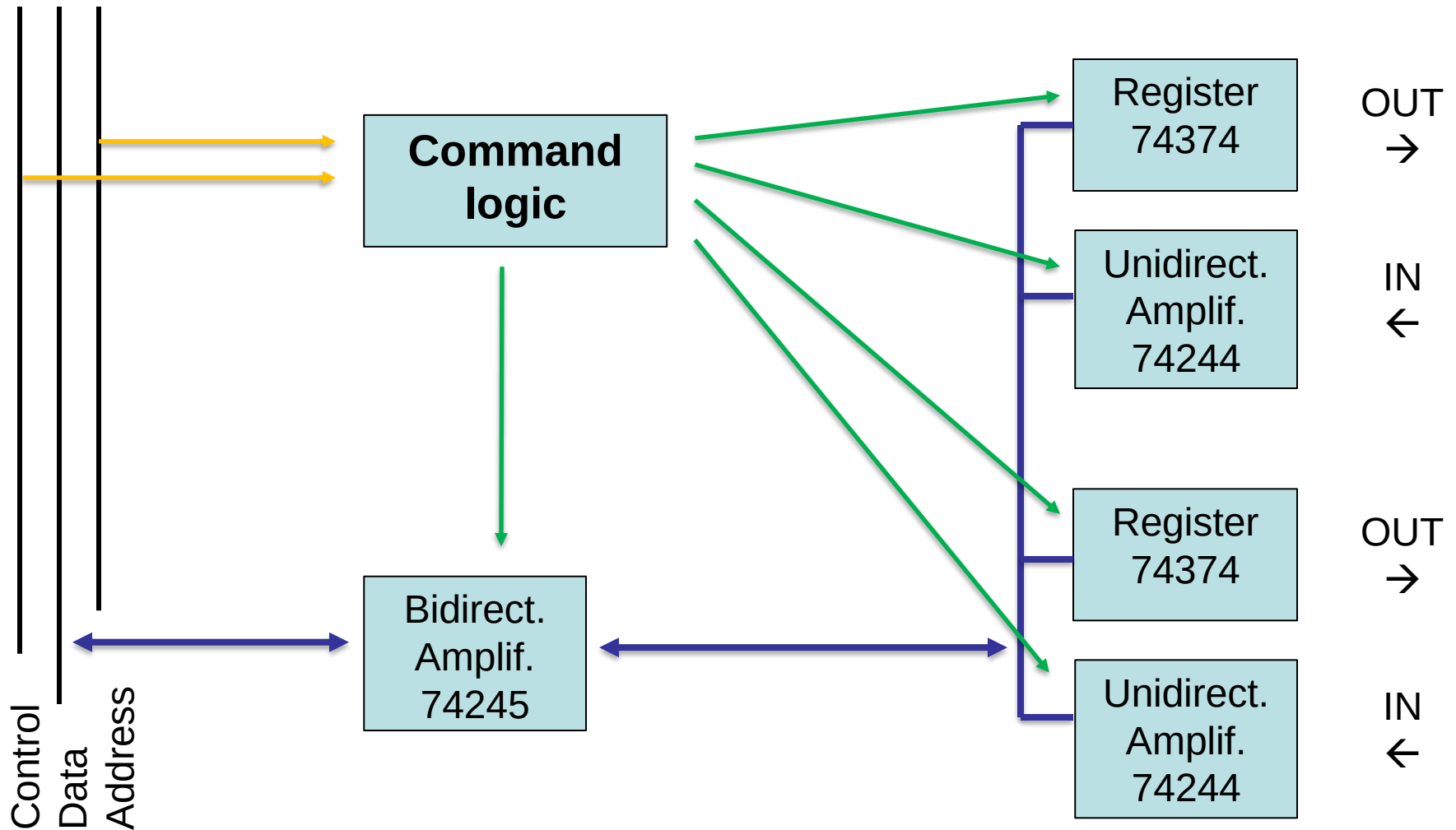
$$V_{OH} = \frac{V_{CC\min} R_2}{R_1 + R_2} \approx 3,5V$$



- Design an input / output interface that loads the bus of a microsystem with an HCT load and has the following ports: 1 input port and an output port at the FE_H address, 1 input port and an output port at the FD_H address. Linear selection is used to activate the ports. The microsystem bus has 8 address lines, 8 data lines and two command signals for reading and writing respectively (RD \, WR \).



- 2 Input ports + 2 output ports
- 8 address lines (A7 ... A0)
- 8 data lines (D7 ... D0)
- 2 command signals (RD\, WR\)

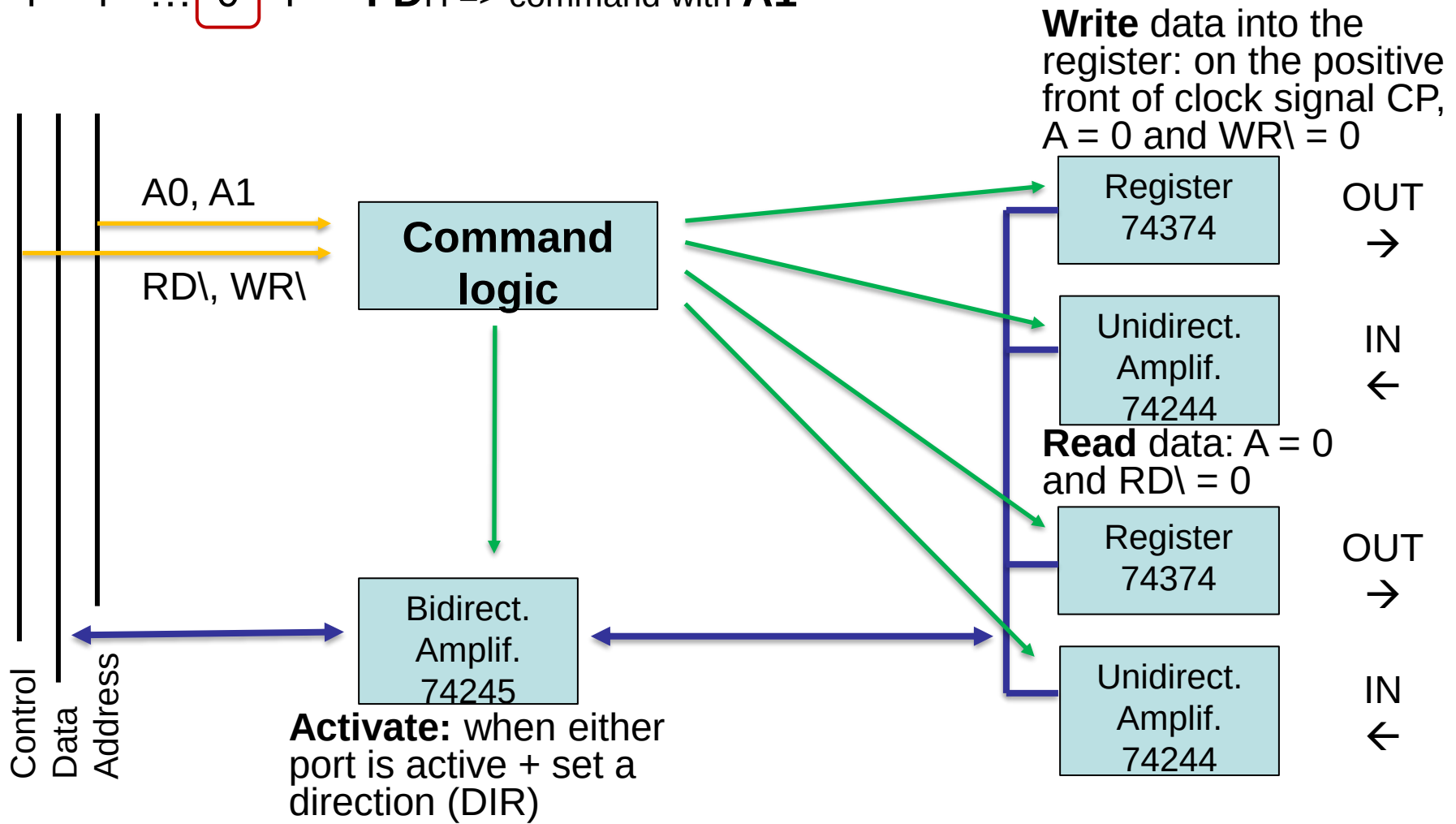


- 1 input port + 1 output port FEH
- 1 input port + 1 output port FDH

A7 A6 ... A1 A0

1 1 ... 1 0 = FEH => command with **A0**

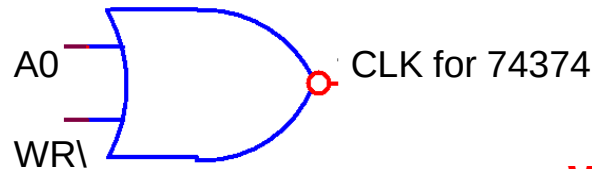
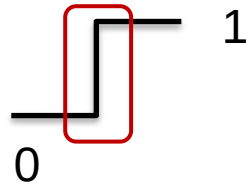
1 1 ... 0 1 = FDH => command with **A1**



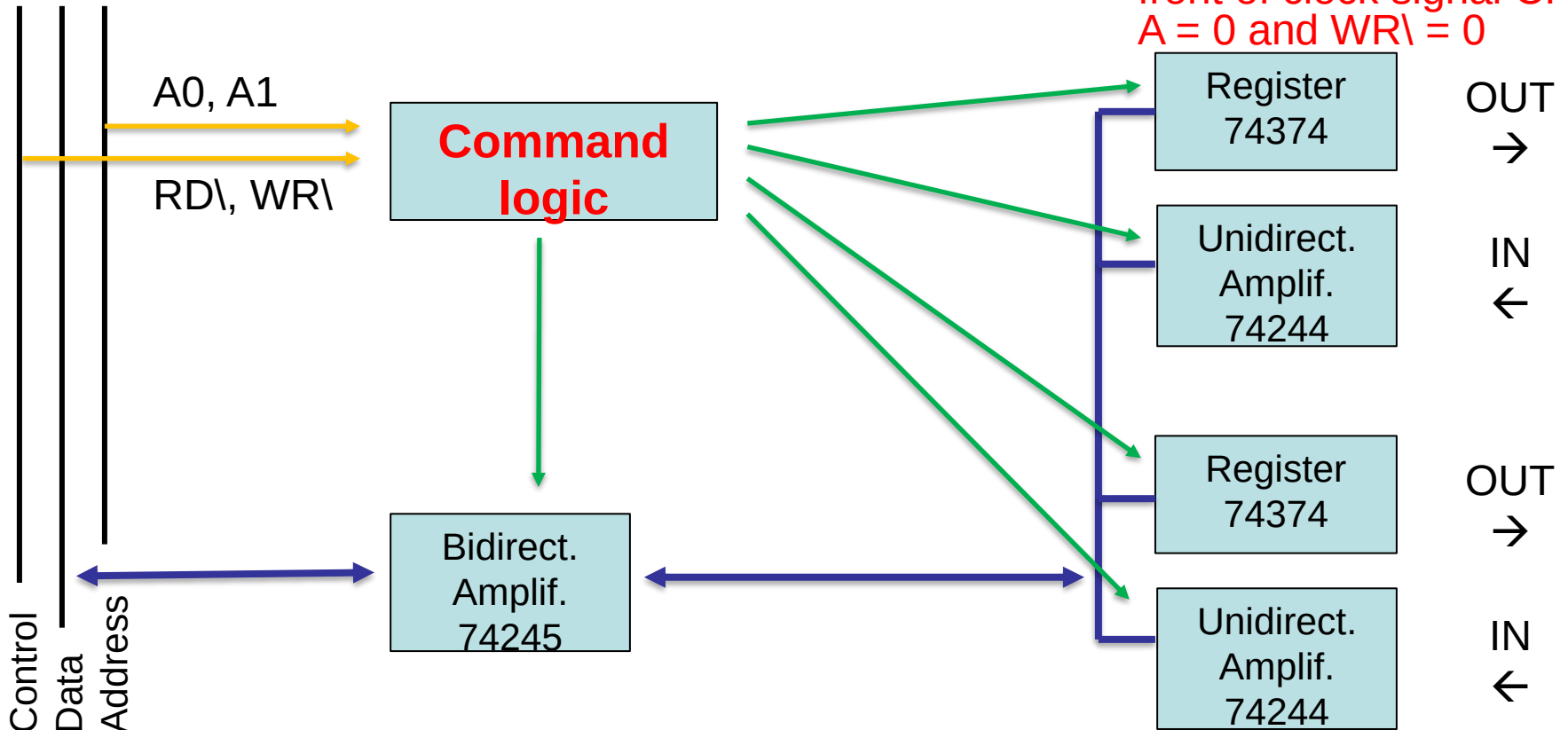
Write (first port): $A0 = 0$, $WR\ = 0$, positive front

NOR gate gives us the needed transition from 0 to 1

A0	WR\	out
0	0	1
0	1	0
1	0	0
1	1	0



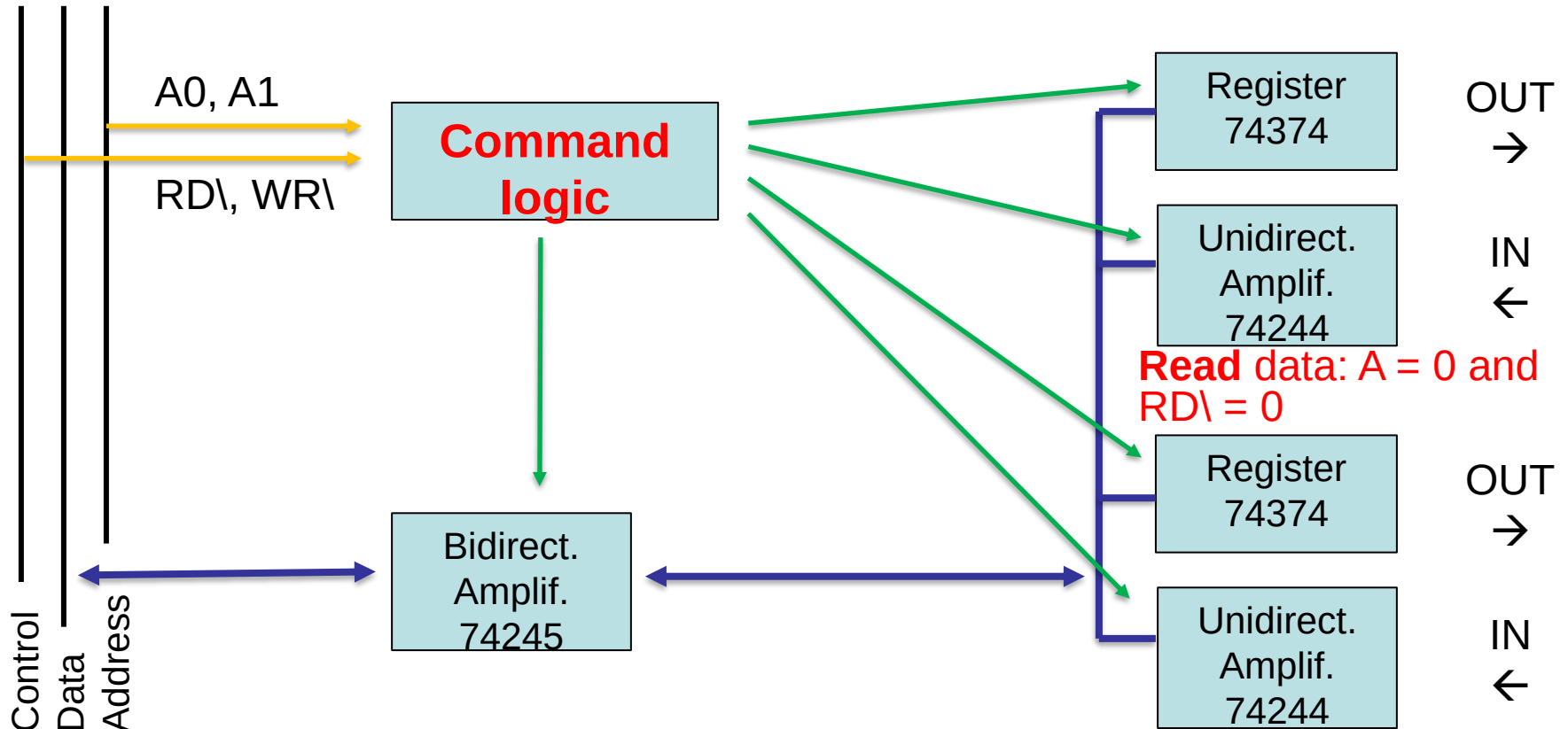
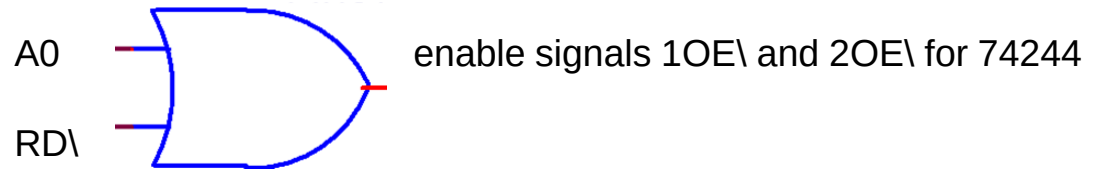
Write data into the register: on the positive front of clock signal CP, $A = 0$ and $WR\ = 0$



Read (first port): $A0 = 0, RD\ = 0$

OR gate gives us the needed output only when both signals are 0

A0	RD\	out
0	0	0
0	1	1
1	0	1
1	1	1



Enable 74245 (OE\=0) and set the direction (DIR)

- If DIR = 0 then direction from B -> A (read the input port and send data to the data bus)
- If DIR = 1 then direction from A -> B (read data bus and send data to output port)

AND gate

0	0	0
0	1	0
1	0	0
1	1	1

OR gates

NOR gates

OR gates

that drive the input ports
(if one **input** port is
active then DIR = 0; else DIR =1 meaning

enable signal for 74245

DIR signal for 74245

output port)

